

Surface hydroxyl-induced Pt₀ clusters on TiO₂ for synergistic water gas shift catalysis

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

This manuscript addresses the role of surface hydroxyls in modulating the catalytic behavior of Pt/TiO₂ systems for the water-gas shift (WGS) reaction. While the overarching topic is of high relevance, particularly regarding metal-support interactions and dynamic surface chemistry, the manuscript contains several conceptual oversimplifications, grammatical issues, and unsupported conclusions that must be addressed before publication.

1. Conceptual Framing and Literature Context

"Supported metal-oxide catalysts have garnered attention due to their ability to form efficient catalytic interfaces through metal-support interactions, where the active metal and the support collectively activate different reactant molecules" This opening sentence suggests that the primary reason for using supported metal catalysts is to facilitate bifunctional activation (e.g., metal vs. support activating distinct molecules). This is misleading. The dominant reason for employing oxide supports is typically related to dispersion and thermal/mechanical stability, not inherently bifunctional activation. Furthermore, reference 8 does not support this bifunctional claim.

"Pt/TiO₂, as a typical catalyst"

What is meant by "typical catalyst"? This phrasing is vague and should be clarified. Typical in what context? WGS? Hydrocarbon reforming?

"which frequently is the rate-determining step of the WGS reaction"

Please provide references to support this assertion regarding the RDS of WGS.

"the electron-deficient Pt sites formed in the Pt^{δ+}-O_v-Ti³⁺ structure accordingly due to the electronic metal-support interaction (EMSI)"

The term "EMSI" must be defined more clearly. What is the specific electron transfer or orbital overlap occurring here? Why invoke a formal term like EMSI without clarifying the mechanism or providing a reference?

"exhibit the inadequate adsorption capacity for CO molecules, leading to limited WGS reaction activity"

If so, isn't that the role of the Pt nanoparticles themselves? Please reconcile the contradiction here: if Pt^{δ+} is inactive, why does that affect the role of Pt₀ nanoparticles, which are present?

"to match the abundant defect structure"

Clarify what structure is being referred to. It appears to be the TiO₂ support, not the Pt.

"loss of abundant active sites"

Why? Is this a geometrical equivalency argument or an adsorption energy-driven rationale? Please clarify. Also consider size effects in CO adsorption: very small Pt particles do not bind CO efficiently (see <https://doi.org/10.1021/ja058282w>).

"provides a guarantee for preserving the size of the"

This is too strong. No treatment "guarantees" particle size preservation.

"on TiO₂ facilitate hydrogen spillover has been clearly proposed^{35,36}."

Please clarify the direction of hydrogen spillover.

"is capable of directly reacting with CO adsorbed on Pt⁰ sites."

Grammar: "are capable"

"precise regulation of the hydroxyl group content is essential, as their effectiveness is not always positively correlated with their content^{40,41}."

Why? What are the competing mechanisms that reduce catalytic effectiveness at high hydroxyl coverage?

"The obtained electron-rich Pt⁰ species was revealed"

Grammar: "were revealed"

"hydroxyl groups during the subsequent reaction acted as active intermediates"

Adsorption and reaction intermediacy is not the same as catalytic activity. Please do not conflate.

"This work achieved the comprehensive enhancement in the activation capability of reactant molecules in"

How much? Please be quantitative. What is the observed improvement? Compared to what literature benchmarks?

"hydroxyl content on the support to obtain the target Pt/TiO₂ catalyst was shown in Figure 1a."

Grammar: "is shown"

2. Language and Clarity

The introduction is relatively well-written, but the remainder of the manuscript suffers from frequent grammatical errors and awkward phrasing. Just 2 quick examples:

- "Commercial anatase (denoted as TiO₂-A) as precursor was hydroxylated by... which denoted as".
- "The surface structure of modified TiO₂ and its effects on the state of supported Pt species need to be clearly characterized".

3. Experimental Issues and Data Interpretation

FTIR Spectra:

- The materials appear extremely wet in the FTIR results in Fig S2. All samples should be dried (preferably >100 °C) prior to spectral acquisition to allow proper comparison of hydroxyl features and FTIR spectra for all samples should be shown.

Claims of Pt Encapsulation:

- The conclusion that Pt is "embedded" in the TiO₂ lattice is based on a single triangular particle image (Supplementary Figs. S3, S4). This is insufficient. Support this claim with catalytic activity suppression or CO chemisorption data to show surface-area loss.
- Figure S4 only presents a narrow IR range. No comparison with other samples is shown. The term "used" is ambiguous; please clarify whether this means spent catalyst.

Figure 1d – Volcano Plot:

- 4 data points are insufficient to define a volcano curve. Error bars are missing (everywhere in the article; please add). Lines should be described in the caption and justified physically.

Redundancy and Missing Information:

- Figure 1f is redundant with 1d.
- Figure 1e should show TGA of samples treated at 200 and 800 °C.
- Figure S7 does not convincingly show a significant difference in OH groups. The conclusion of "significantly more abundant" is not justified.

Particle Size and Dispersion:

- No average particle size data are provided for any sample in Figure 1. If particle size changes, so will activity, regardless of hydroxyl concentration.
- Provide HAADF-STEM with statistical analysis (>100 particles), histograms, and support with chemisorption data to report surface exposure.
- Report TOF values.

TPR Analysis:

- The H₂-TPR data (Fig. 2a) do not prove hydrogen spillover. Lower reduction temperatures can also arise from water loss or hydroxyl dehydrogenation. FTIR suggests the material is wetter, not necessarily more reducible.
- Assigning changes in reducibility to Pt species alone (e.g., Pt²⁺ → Pt⁰) without structural or XAS data is speculative.

XPS Data:

- The XPS spectra (Fig. 1c) show minimal differences. Claims of electron-rich Pt⁰ formation are not distinguishable from noise. Additional XAS data are needed.

CO Adsorption / DRIFTS Interpretation:

- The assignment of bands to Pt⁰ vs. Pt^{δ+} is highly tentative.
- CO-TPD experiments only support stronger binding, not overall "adsorption capability."

"CO adsorbed on the electron-deficient $\text{Pt}^{\delta+}$ species"
Again, this assignment rests on uncertain XPS data.

"In contrast, for the 0.3Pt/(TiO₂-T-500) catalyst... CO₂ products... Fig. 2i"

This claim is highly questionable. Figure 2h shows a CO₂ peak for the "A" sample that is larger than that for the "T-500" sample. The conclusion that CO₂ is not formed on the A sample is unsubstantiated. Under Ar, no desorption curves are provided for CO, which complicates interpretation. It is also very plausible that the T-500 sample simply adsorbs CO more strongly due to particle size differences.

Reaction Mechanism Claims:

"This corresponded to the conclusion that oxygen vacancies were formed by the consumption of surface hydroxyls by CO during the reaction..."

Again, this could also be due to differences in particle size or suboxide shell formation. Particle sizes must be reported for all fresh and spent catalysts.

Overall

This study explores a timely and important topic, but currently suffers from overinterpretation of spectroscopic data, lack of rigorous controls, and incomplete physical characterization. Stronger claims about site-specific activity or electronic structure require corroboration by XAS, chemisorption, and particle size analysis. Many conclusions are not fully supported by the presented data.

Substantial revisions are necessary before this manuscript can again be considered for publication in Nature Communications.

Reviewer #2

(Remarks to the Author)

Jia and coauthors present a compelling strategy for enhancing the catalytic performance of supported metal-oxide catalysts by modulating the electronic structure of active sites through surface hydroxyl groups. This work convincingly demonstrates the formation of electron-rich Pt^0 clusters (~2 nm) on hydroxylated TiO₂, in contrast to conventional $\text{Pt}^{\delta+}$ -Ov-Ti³⁺ interfacial structures. This modification not only enhances CO adsorption and activation but also couples with hydroxyl-assisted oxygen vacancy formation to promote H₂O dissociation, leading to synergistic improvement in the water-gas shift (WGS) reaction. The mechanistic insights provided into how surface hydroxyl groups contribute both to the electronic modulation of Pt and the dynamic restructuring of the oxide support are novel and significant. I find the study well-structured and impactful; however, I suggest the authors provide additional experimental or theoretical validation of the proposed Pt^0 -hydroxyl interaction mechanism and quantify the contribution of each step (CO adsorption vs. H₂O dissociation) to the overall WGS enhancement.

1. Why is there such a large shift (~0.7 eV) in the O1s binding energy of lattice oxygen (O_{latt}) in TiO₂-T-UC compared to TiO₂-T-500 and TiO₂-A? Please explain the origin of this significant difference.
2. The statement "However, excessive surface hydroxyls caused the supported Pt species to be completely embedded in the TiO₂ support lattice during the WGS reaction, resulting in the loss of active metal sites (Supplementary Fig. S3 and S4)" requires quantitative data or literature support. Only two STEM images alone are not sufficient to support such a conclusion. A statistical comparison of the post-reaction STEM data between TiO₂-T-500 and the hydroxyl-rich sample would help clarify how the OH content influences the embedding or migration of Pt species.
3. The frequent mention of "OH content of TiO₂-T-500 (1.3 mmol g⁻¹)" gives the impression that the value may have been selectively emphasized. Please consider adjusting Figure 1 to avoid repetitive or potentially misleading data presentation.
4. Please provide the WGS reaction conditions in the legend of Figure 1, especially those corresponding to Figure 1d. This information is necessary for clarity and reproducibility.
5. The interpretation of Pt 4f spectra in Figure 2c requires further clarification. Specifically, 0.3Pt/(TiO₂-T-500) predominantly shows metallic Pt^0 species, while 0.3Pt/(TiO₂-A) mainly exhibits Pt^{2+} or Pt^{4+} states. The orange peaks in both XPS spectra are currently labeled as Pt^0 , but the binding energy difference between them (~0.7 eV) is too large to be attributed to the same oxidation state. Please clarify the specific assignments of each deconvoluted peak.
6. The phrase "the formation of electron-rich Pt^0 species" is somewhat inappropriate. It may be unnecessary to emphasize "electron-rich," as the main point is the formation of metallic Pt^0 species. There is no clear evidence that these Pt^0 species are more electron-enriched than typical Pt^0 , as indicated by the XPS results in Figure 2c.
7. In the in situ DRIFTS spectra of the 0.3Pt/(TiO₂-A) catalyst, why does the CO₂ signal appear at the very beginning (Figure S14), while for 0.3Pt/(TiO₂-T-500) the signal is much weaker or absent? This suggests that the $\text{Pt}^{\delta+}$ -OH-Ti⁴⁺ interfacial sites in the former are highly reactive initially, rapidly oxidizing CO to CO₂. Please discuss this. Also, how should we interpret the volcano-shaped curve of CO₂ formation in Figure 2i? A clear explanation is needed in the text.
8. As the authors state, "These results revealed that the electron-rich Pt^0 closer to the interface was more active in activating CO molecules." However, is it possible that the " Pt^0 near the interface" is in fact the $\text{Pt}^{\delta+}$ species located at the $\text{Pt}^{\delta+}$ -OH-Ti⁴⁺ interface, rather than fully metallic Pt^0 ? This may involve dynamic transformation between $\text{Pt}^{\delta+}$ and Pt^0 driven by OH consumption and water dissociation during the reaction. Consider using DFT simulations and valence state calculations to further support this hypothesis.
9. In Figure 3b, the CO₂ and H₂ signals during the second CO injection are significantly weaker than those during the first injection. This indicates that the interfacial OH species regenerated via water dissociation are less abundant than the native surface OH species originally present on the catalyst. What could be the reason? Is it due to structural changes in the catalyst or sintering of the Pt nanoparticles?
10. After the long-term stability test, how does the Pt 4f XPS spectrum of the 0.3Pt/(TiO₂-T-500) catalyst compare with the

fresh sample? Are there any notable changes in the oxidation state of Pt species?

11. Based on the above concerns, I would like to recommend the publication of this work in Nature Communications.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors have revised their manuscript and the central idea remains that tuning the hydroxyl population on TiO₂ has two coupled effects: it biases reduced Pt toward a more metallic electronic character that strengthens CO binding and activation, and it promotes oxygen-vacancy generation during turnover in a way that enhances water activation. The overall narrative is coherent and, at a high level, plausible for WGS on reducible oxides. I also find the materials handle conceptually appealing and potentially useful: deliberate hydroxyl regulation via a titanate-derived route and controlled calcination to reach an apparent optimum hydroxyl content at 500 °C. The optimization itself is one of the more convincing aspects of the work, particularly because the authors acknowledge that excessive hydroxylation can lead to Pt encapsulation in the uncalcined nanotubular precursor and they relate that state to suppressed CO uptake and suppressed activity for the hydroxyl-rich TiO₂-T-UC sample, rather than relying on microscopy alone.

The revision is strengthened by the addition of TOFs, chemisorption-derived dispersions, reaction orders, and additional spectroscopy. The reported performance differences are nontrivial and persist after normalization to accessible Pt sites as the authors define them. In particular, the revised manuscript reports a rate of 62.7 molCO gPt⁻¹ h⁻¹ and a TOF of 6.15 s⁻¹ for 0.3Pt/(TiO₂-T-500) at 250 °C, compared with 31.4 molCO gPt⁻¹ h⁻¹ and 3.54 s⁻¹ for 0.3Pt/(TiO₂-A). The authors also state that the two catalysts show similar apparent activation energies and therefore likely follow the same overall mechanism, which is an important and appropriate framing that supports a rate-enhancement claim rather than a claim of a fundamentally new pathway.

My main concern is that key mechanistic statements still read more certain than the current evidence warrants. In several places the manuscript relies on chains of observations that are mutually consistent but not uniquely discriminating, and the conclusions drift from “consistent with” toward “demonstrated”. This is most evident in the assignment of Pt⁰ versus Pt^{δ+}. The assignment is still based primarily on CO-DRIFTS band positions and small XPS shifts, supported by an in situ Pt L₃-edge XANES comparison acquired under H₂ pretreatment. The authors themselves note limitations here, including signal-to-noise constraints at 0.3 wt% and the fact that the measurement conditions are not operando WGS. The chemical rationale offered for the DRIFTS shift (reduced back-donation on electron-deficient Pt^{δ+} leading to higher CO stretching frequency) is reasonable, but it does not resolve the core ambiguity: on Pt/TiO₂, CO band positions can be influenced by oxidation state as well as particle size, coverage, ensemble effects, and interfacial geometry. The authors have added post-reaction HAADF-STEM statistics and CO chemisorption on spent catalysts, and the mean sizes appear similar after reaction (about 1.6–1.7 nm). However, the chemisorption table shows dispersions that are not identical (about 55.2% for TiO₂-T-500 versus 48.1% for TiO₂-A in the provided dataset). With such low metal loading, this difference is not obviously negligible, and it matters because the chemisorption-derived site count is being used both to normalize activity (TOF) and to support mechanistic interpretation. If the claim is that intrinsic site quality changes, the manuscript would benefit from a clearer demonstration that CO uptake and dispersion estimates are robust and not systematically biased by support-dependent spillover, readsorption, or differences in CO binding strength that could affect the inferred site count. At minimum, the authors should temper conclusions that hinge critically on dispersion-derived normalization and explicitly discuss the uncertainty and assumptions.

The second major mechanistic pillar, that hydroxyl consumption by CO generates oxygen vacancies that become the dominant loci for water dissociation, is intriguing and not unreasonable, but it is still somewhat over-interpreted. The revision links vacancy formation to EPR changes after WGS and to Ti-related spectral features interpreted as vacancy signatures, and then links “more vacancies” to “better water activation” through reaction orders and a kinetic isotope effect. The reported KIE of roughly 1.65 upon substituting D₂O for H₂O indicates that O–H activation is kinetically relevant, but it does not uniquely identify the elementary step, nor does it establish that vacancies created specifically by hydroxyl consumption are the operative active sites for that step. Moreover, the manuscript also states that hydroxyl signals increase after reaction due to water dissociation and adsorption, implying that hydroxyls are continuously regenerated during WGS. In such a dynamic system it is important to distinguish hydroxyls as spectators, hydroxyls as true intermediates on the main catalytic cycle, and hydroxyls as part of a parallel redox process that modulates defect populations without being kinetically dominant. At present the evidence is consistent with the proposed picture, but causality remains incompletely established. I would therefore ask for at least one genuinely discriminating experiment or, alternatively, a more conservative mechanistic framing. Suitable discriminating experiments could include isotope tracing (for example H₂¹⁸O, C¹⁸O, or ¹⁸O-enriched supports) to determine whether oxygen in CO₂ derives predominantly from water, lattice oxygen, or hydroxyl oxygen under the relevant catalyst states, or operando measurements that correlate a quantitative defect metric with instantaneous rate under controlled perturbations.

A further concern is a plausible confound that is not convincingly excluded. The hydroxylated support is produced using concentrated NaOH and acid washing to form titanate intermediates before calcination. Trace residual sodium or chloride can measurably alter TiO₂ surface chemistry, hydroxyl stability, vacancy formation energetics, and WGS kinetics, and these effects could mimic a “hydroxyl regulation” trend. Given that the manuscript compares performance against literature benchmarks that include alkali-modified Pt/TiO₂ systems, it becomes especially important to demonstrate that residual promoters are not contributing to the observed behavior. If not already provided, elemental analysis such as ICP/OES (or similarly convincing quantification) for Na and Cl on the final catalysts should be included, along with a brief discussion of detection limits.

I would also like to comment on the balance of the cited literature. Approximately half of the references in the manuscript are authored exclusively by Chinese research groups, which is unusually high for a general catalysis paper in Nature Communications. Many of these citations are used to support broad and well-established concepts such as metal–support interactions, hydroxyl effects, and defect-mediated water activation. At the same time, several foundational and widely recognized contributions from the broader international literature are under-represented or absent. This includes classical frameworks for metal–support interaction and reducible oxide chemistry, as well as established kinetic and mechanistic studies of WGS on Pt/TiO₂. The current citation pattern risks giving the impression that recent regional literature defines these concepts, rather than placing the present work within the full historical and international context of the field. Such a balance is problematic from a scholarly perspective, and I would strongly recommend that the author balance the references to better reflect the breadth of prior work and to clearly acknowledge the foundational studies on which this manuscript builds.

I would also like to comment on a conceptual statement made in the rebuttal, which appears to carry over implicitly into the revised mechanistic framing. The authors state that “for gas-phase reactions involving two distinct reactant molecules, such as the water–gas shift reaction, catalysis at a single type of active site is often insufficient.” As written, this statement remains misleading and introduces conceptual ambiguities that has not yet been fully resolved in the revised text. The overwhelming majority of heterogeneous catalytic reactions involve two or more reactants, and yet in many well-established systems the dominant chemistry occurs on metal sites alone, even when multiple elementary steps and adsorbates are involved. Moreover, supported metal nanoparticles do not constitute a “single type of active site” in any strict sense, as they inherently present a distribution of terraces, edges, corners, and metal–support interfacial sites. While bifunctional or metal–support cooperative mechanisms certainly exist and are well documented for some reactions and materials, they are not the general rule, and invoking them requires careful evidence rather than broad generalization. I would therefore encourage the authors to revise the language in both the manuscript and the rebuttal to avoid implying that multi-reactant gas-phase catalysis generally requires chemically distinct active sites on different phases. A more accurate framing would acknowledge that metal–support cooperation is one possible mechanistic motif for WGS on reducible oxides, but not an inherent requirement dictated simply by the presence of multiple reactants.

Finally, the manuscript's positioning still risks overselling novelty in a field where the coupled roles of hydroxyls, defect chemistry, and metal electronic structure in WGS have been discussed extensively. The work will be strongest if it is positioned around the specific and defensible contribution: a practical hydroxyl-regulation approach that yields a reproducible optimum and correlates with improved activity, with a mechanistic interpretation that is consistent with multiple probes but not asserted as uniquely proven where it is not. In particular, the manuscript should avoid implying that Pt L₃-edge data under H₂ pretreatment definitively establishes Pt surface oxidation state under WGS conditions.

Overall, the revision addresses several legitimate concerns and the performance improvements, including TOF-normalized differences as presented, are significant. I believe the work can be suitable for Nature Communications provided that the mechanistic claims are brought into tighter alignment with what is strictly demonstrated, and that either a discriminating experiment is added to strengthen the hydroxyl-to-vacancy-to-water-activation causality or the mechanistic framing is thoroughly revised to be appropriately cautious. In addition, the manuscript should pre-emptively address residual-ion effects from the synthesis route with quantitative elemental analysis or equivalent evidence.

Reviewer #2

(Remarks to the Author)

The authors have addressed all of my concerns, and I recommend the publication of this work in Nature Communications.

Version 2:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors have now responded appropriately and substantively and have largely addressed my concerns. Small residual things remain. E.g., the manuscript still contains a few sentences that read more definitive than the new caveats warrant—for example, language like “predominant Pt⁰ species were present...” and “suggested... in the absence of size effects” is still stronger than “consistent with”, given their own acknowledgement that CO-band positions can shift with coverage/geometry - but I believe this is acceptable.

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Responses to the Reviewers' Comments and the Corresponding Revisions

Reviewer #1:

Comment: *This manuscript addresses the role of surface hydroxyls in modulating the catalytic behavior of Pt/TiO₂ systems for the water-gas shift (WGS) reaction. While the overarching topic is of high relevance, particularly regarding metal-support interactions and dynamic surface chemistry, the manuscript contains several conceptual oversimplifications, grammatical issues, and unsupported conclusions that must be addressed before publication.*

Response: We sincerely appreciate the reviewer for the careful reading of the manuscript and the constructive comments, which have greatly helped us improve the quality of the work. We have carefully revised the entire manuscript and provided more comprehensive evidence. We hope this revised version can be accepted to *Nature Communications*.

1. Conceptual Framing and Literature Context

Comment 1: *"Supported metal-oxide catalysts have garnered attention due to their ability to form efficient catalytic interfaces through metal-support interactions, where the active metal and the support collectively activate different reactant molecules"*

This opening sentence suggests that the primary reason for using supported metal catalysts is to facilitate bifunctional activation (e.g., metal vs. support activating distinct molecules). This is misleading. The dominant reason for employing oxide supports is typically related to dispersion and thermal/mechanical stability, not inherently bifunctional activation. Furthermore, reference 8 does not support this bifunctional claim.

Response: We sincerely appreciate the professional comments of the reviewer. Indeed, as the reviewer rightly pointed out, **supported metal–oxide catalysts have attracted extensive attention owing to their high dispersion and stability of metals. This**

viewpoint has been revised and showed on Page 3, lines 5–6 in the revised manuscript (highlight in yellow).

Moreover, for gas-phase reactions involving two distinct reactant molecules, such as the water gas shift (WGS) reaction, catalysis at a single type of active site is often insufficient. The synergistic effect of multiple catalytic sites is required to facilitate the reaction. In supported metal-oxide catalysts, when the reducible oxide serves as the support, the synergistic effect between the active metal and oxygen vacancies becomes crucial. Shen et al. investigated the realistic structure of the Cu/CeO₂ catalyst and its structure-activity relationship in the WGS reaction. In their study, Cu clusters provide Cu⁺ sites for effective CO adsorption, while oxygen vacancies on the CeO₂ surface dissociate H₂O into hydroxyl species (Chen, A. et al. *Nat. Catal.* **2019**, 2, 334–341). Frenkel et al. also demonstrated that the Pt⁰-O_v-Ce³⁺ structure constitutes the vital active site for the WGS reaction, where CO adsorption and H₂O dissociation occur at the Pt⁰ sites and oxygen vacancies, respectively (Li, Y. et al. *Nat. Common.* **2021**, 12, 914). In our work, Pt⁰ sites in the clusters function to adsorb CO, while a high density of oxygen vacancies formed in situ during the reaction efficiently cleaves H₂O. This similarly underscores the important role of synergistic sites in the activation of reactant molecules within supported metal-oxide catalysts. **Therefore, we have added the corresponding description on Page 3, lines 6–7 (highlight in yellow), with citations to the aforementioned literature as 3 and 4.**

Comment 2: "Pt/TiO₂, as a typical catalyst"

What is meant by "typical catalyst"? This phrasing is vague and should be clarified. Typical in what context? WGS? Hydrocarbon reforming?

Response: We are grateful to the reviewer for the constructive comment. **Pt/TiO₂ system stands as an exemplary catalytic system that has been extensively investigated in a variety of pivotal reactions, including the water gas shift reaction, CO oxidation, hydrogenation, and hydrocarbon reforming (Iida, H. et al. *Appl.***

Catal. A Gen. **2006**, 298, 152–160; Zhang, Y. et al. *ACS Catal.* **2023**, 13, 392–399; Gao, M. et al. *ACS Cent. Sci.* **2022**, 8, 1633–1645; Zhao, H. et al. *ACS Appl. Mater. Interfaces* **2018**, 10, 36954–36960; Chen, J. et al. *Nat. Commun.* **2023**, 14, 1–12; Hu, X. et al. *Angew. Chem. Int. Ed.* **2025**, 64, e202419103; He, C. et al. *Angew. Chem. Int. Ed.* 2024, **136**, e202412308). **We have revised the manuscript on Page 3, lines 8–10 (highlight in yellow). The relevant references have been cited as 5–8.**

Comment 3: "which frequently is the rate-determining step of the WGS reaction"

Please provide references to support this assertion regarding the RDS of WGS.

Response: We thank the reviewer for the valuable suggestion. Accordingly, we have now cited three references (Mavrikakis, M. et al. *J. Am. Chem. Soc.* **2008**, 130, 1402–1414; Odriozola, J. A. et al. *ACS Catal.* **2024**, 14, 14947–14957; Wei, M. et al. *ACS Catal.* **2019**, 9, 2707–2717.), which provide compelling evidence from both DFT calculations and experiments, confirming that H₂O dissociation acts as the rate-determining step in the water gas shift reaction. **The relevant references have been cited as 11–13.**

Comment 4: "the electron-deficient Pt sites formed in the Pt^{δ+}-O_v-Ti³⁺ structure accordingly due to the electronic metal-support interaction (EMSI)"

The term "EMSI" must be defined more clearly. What is the specific electron transfer or orbital overlap occurring here? Why invoke a formal term like EMSI without clarifying the mechanism or providing a reference?

Response: We sincerely thank the reviewer for this insightful comment. The electronic metal-support interaction (EMSI) primarily refers to the electron transfer between the support and the supported metal. The formation of the Pt^{δ+}-O_v-Ti³⁺ structure in Pt/TiO₂ systems involves electron transfer from Pt to TiO₂, thereby generating electron-

deficient $\text{Pt}^{\delta+}$ species. The electron transfer induced by EMSI has been considered as an established phenomenon in the literatures (Dong, Y. et al. *ACS Cent. Sci.* **2022**, 8, 1633–1645; Jiang, J. et al. *Angew. Chem. Int. Ed.* **2025**, 64, e202419103; Yang, Q. et al. *Common. Chem.* **2021**, 4, 54).

The corresponding description has been revised on Page 3, lines 10–12 (highlight in yellow). We have cited these works as references 6, 9, and 10. Once again, we thank the reviewer for the professional and thorough suggestions.

***Comment 5:** "exhibit the inadequate adsorption capacity for CO molecules, leading to limited WGS reaction activity"*

If so, isn't that the role of the Pt nanoparticles themselves? Please reconcile the contradiction here: if $\text{Pt}^{\delta+}$ is inactive, why does that affect the role of Pt^0 nanoparticles, which are present?

Response: We thank the reviewers for the constructive comments. In Pt/TiO₂-based catalyst systems, it is generally accepted that the interfacial sites serve as the active centers for the reaction (Gao, M. et al. *ACS Cent. Sci.* **2022**, 8, 1633–1645; Xiong, H. et al. *J. Am. Chem. Soc.* **2025**, 147, 38204–38214.). For nanoparticle catalysts, Pt^0 species that are distant from the interface are generally catalytically inert and function primarily as spectators during the reaction. Moreover, the electronic structure of the interfacial sites often plays a key role in determining the catalytic performance. However, due to the electronic metal–support interaction (EMSI) between Pt and TiO₂, electrons at the interface tend to transfer from Pt species to TiO₂, resulting in the formation of $\text{Pt}^{\delta+}\text{--O}_\text{v}\text{--Ti}^{3+}$ interfacial sites. Although such interfacial sites facilitate the formation of unsaturated Ti^{3+} and oxygen vacancies, thereby enhancing H₂O dissociation, the electron-deficient $\text{Pt}^{\delta+}$ species exhibit limited ability for CO activation. This limitation hinders effective matching with the enhanced H₂O dissociation capability, thus restricting further improvement in the WGS reaction activity. Therefore, it is essential to modulate the electronic structure of Pt species at the interface. This

would enable the formation of $\text{O}_\text{v}\text{-Ti}^{3+}$ sites to improve H_2O dissociation, while simultaneously maintaining the interfacial Pt species in the metallic state, thereby facilitating more effective activation of CO molecules and synergistically enhancing the WGS reaction activity.

In summary, the insufficient activation of CO by the interfacial $\text{Pt}^{\delta+}$ species does not affect the Pt^0 sites located away from the interface, which typically act as spectators. Rather, our objective is to strategically modulate the structure of the interfacial sites, which are recognized as the active centers, to further enhance their CO activation capability. This would allow it to better match the high H_2O dissociation activity, thereby maximizing the overall performance of the WGS reaction. **The corresponding description has been revised on Page 3, lines 12–16 in the revised manuscript (highlight in yellow).**

Comment 6: "to match the abundant defect structure"

Clarify what structure is being referred to. It appears to be the TiO_2 support, not the Pt.

Response: We appreciate the reviewer for the careful comments. Researchers focus on engineering abundant defect structures on the TiO_2 support surface. To provide greater specificity, the relevant discussion has been updated to “**to match the abundant defect structure on the TiO_2 surface**” on Page 3, lines 18–19 in the revised manuscript (highlight in yellow).

Comment 7: "loss of abundant active sites"

Why? Is this a geometrical equivalency argument or an adsorption energy-driven rationale? Please clarify. Also consider size effects in CO adsorption: very small Pt particles do not bind CO efficiently (see <https://doi.org/10.1021/ja058282w>).

Response: We are grateful to the reviewer for the insightful comment, which helps to clarify the underlying mechanism. The loss of active sites with increased particle size

is primarily a geometrical equivalency argument (Ge, S. et al. *ACS Catal.* **2024**, 14, 2532-2544). The primary reason is the decrease in the dispersion and the specific surface area of the Pt species as their size increases. Smaller nanoparticles have a higher proportion of their total atoms exposed on the surface, which can function as active sites. As the particles grow, a larger fraction of atoms resides in the particle bulk, becoming inaccessible for catalysis, which is regarded as the underlying cause of abundant active site loss. Certainly, as the reviewer rightly points out, catalytic activity is not monotonically increased with decreasing particle size. When the metal species become extremely small (e.g., < 1 nm), the formation of positively charged metal sites is promoted (Leendert Bezemer, G. et al. *J. Am. Chem. Soc.* **2006**, 128, 3956-3964). This electron deficiency suppresses the crucial π -backdonation from the metal d-orbitals to the $2\pi^*$ anti-bonding orbital of CO, thereby reducing the strength of the Pt–CO bond and consequently weakening CO adsorption. Therefore, **precise regulation of the electronic structure of Pt sites while controlling the size of metal species is essential yet challenging. The corresponding description has been shown in the revised manuscript on Page 3, lines 21–26 (highlight in yellow). The corresponding references have been cited as 15 and 16.**

Comment 8: "provides a guarantee for preserving the size of the"

This is too strong. No treatment "guarantees" particle size preservation.

Response: We highly appreciate the reviewer's valuable comments. We have revised to **"contributes to maintaining the size of the active metals"** on Page 4, lines 2–3 (highlight in yellow).

Comment 9: "on TiO₂ facilitate hydrogen spillover has been clearly proposed^{35,36}."

Please clarify the direction of hydrogen spillover.

Response: We thank the reviewer for the insightful comment. We have clarified that **the direction of hydrogen spillover is from the metal sites to the support surface** and have modified the manuscript accordingly on Page 4, line 4.

Comment 10: "is capable of directly reacting with CO adsorbed on Pt⁰ sites."

Grammar: "are capable"

Response: We appreciate the reviewer's careful comment and apologize for our mistake. **We have corrected it on Page 4, line 8.**

Comment 11: "precise regulation of the hydroxyl group content is essential, as their effectiveness is not always positively correlated with their content^{40,41}."

Why? What are the competing mechanisms that reduce catalytic effectiveness at high hydroxyl coverage?

Response: We thank the reviewer for the constructive comment. In catalysis, the influence of surface hydroxyl groups does not follow a trend of monotonic improvement. An optimal coverage exists, which must be evaluated on a case-by-case basis for different catalytic systems. Firstly, the exposure of active sites can be hindered by strong metal–hydroxyl interactions. For instance, studies on Ru catalysts have shown that the strong affinity for adsorbed hydroxyls poisons the active sites and degrades performance. This issue can be addressed by introducing SnO₂, which competes with Ru for OH adsorption, thereby effectively mitigating the site poisoning (Zhang, J. et al. *Angew. Chem. Int. Ed.* **2022**, 61, e202209486). Moreover, hydroxyl groups can alter the oxidation state and dispersion of the active metal. However, the state of metal species induced by the high hydroxyl coverage is not necessarily the most catalytically active. As demonstrated by the work of Li et al. (Li et al., *J. Am. Chem. Soc.* **2025**, 147, 30326–30338), the content of surface hydroxyls can be tailored to control the size of Ag on Al₂O₃. They found that the high hydroxyl coverage anchors

Ag as single atoms. Consequently, while this single-atom configuration delivered optimal performance for the C₃H₆-SCR reaction, it exhibited negligible activity in both C₂H₄ epoxidation and NH₃-SCO reactions. In contrast, the lower hydroxyl coverage facilitated the formation of Ag nanoparticles, which was proved to be the most active for the latter reactions. This finding clearly illustrates that no positive correlation exists between surface hydroxyl content and catalytic activity, as different reactions require distinct state of active site. Therefore, the hydroxyl content must be strategically tuned to achieve optimal performance for the specific catalytic reaction. **Based on the two points above, we have revised the manuscript on Page 4, line 11–16 (highlight in yellow). And we have cited the aforementioned references as 34 and 35. Once again, we thank the reviewer for the insightful comments.**

Comment 12: "The obtained electron-rich Pt⁰ species was revealed"

Grammar: "were revealed"

Response: We appreciate the reviewer's thorough comment and apologize for the mistake. **The manuscript has been revised accordingly, as indicated on Page 4, line 19.**

Comment 13: "hydroxyl groups during the subsequent reaction acted as active intermediates"

Adsorption and reaction intermediacy is not the same as catalytic activity. Please do not conflate.

Response: We sincerely thank the reviewer for the insightful comment. We agree with the reviewer that strengthened reactant adsorption at the individual site and the presence of reaction intermediates cannot be solely interpreted as direct evidence for enhanced catalytic activity. In our work, the hydroxyl groups constructed on the catalyst surface

not only function as intermediates, reacting with CO to form CO₂, but, more importantly, their consumption during the reaction concurrently generates abundant oxygen vacancies. These vacancies subsequently serve as active sites for H₂O dissociation. This enhanced H₂O dissociation capability synergizes with the strong adsorption and activation of CO molecules on Pt⁰ sites, thereby collectively promoting the observed increase in the reaction rate. **To avoid any confusion, we have revised the relevant text in the revised manuscript on Page 4, lines 20–23 (highlight in yellow).**

Comment 14: "This work achieved the comprehensive enhancement in the activation capability of reactant molecules in"

How much? Please be quantitative. What is the observed improvement? Compared to what literature benchmarks?

Response: We appreciate the reviewer's professional comment. Generally, the improvement in the activation capability of reactant molecules corresponds to the increase in the catalytic reaction rate. The turnover frequency (TOF) defines the intrinsic activity per active site and can serve to quantify the activation capability of the active sites towards reactant molecules. As presented in Table R1-1, the 0.3Pt/(TiO₂-T-500) catalyst exhibited the TOF nearly twice that of the reference 0.3Pt/(TiO₂-A) catalyst. Meanwhile, this TOF value demonstrated the clear advantage over those reported for other Pt-based catalysts in the literature. These results indicated that the 0.3Pt/(TiO₂-T-500) catalyst possess an enhanced capability for activating reactant molecules. In addition, the qualitative assessment of reactant activation capability is also widely employed, where the value of the reaction order can represent the activation capacity for reactant molecules. As shown in Figure R1-1a and b, at both 250 and 290 °C, the H₂O reaction order of the 0.3Pt/(TiO₂-T-500) catalyst showed the 20% reduction compared to the 0.3Pt/(TiO₂-A) reference. This observed reduction suggested the improvement in the H₂O activation. Similarly, the CO reaction order for the 0.3Pt/(TiO₂-T-500) catalyst is also approximately half that of the 0.3Pt/(TiO₂-A)

catalyst at both 250 °C and 220 °C, indicating the enhanced CO activation capability (Figure R1-1c and d).

To more clearly demonstrate the enhanced activation of reactant molecules, we have now quantified this improvement using the reaction rate and TOF value in the revised manuscript. **The text has been revised on Page 4, line 23–26 (highlight in yellow). The Table R1-1 has been shown as Table S4 in supplementary information. The Figure R1-1a has been shown as Figure 3f in the revised manuscript. The Figure R1-1b has been shown as Figure S31 in supplementary information. The Figure R1-1c has been shown as Figure 2g in the revised manuscript. The Figure R1-1d has been shown as Figure S20 in supplementary information.**

Table R1-1. Comparison of reaction rate and TOF values for WGS reaction over various catalysts at 250 °C.

Samples	Rate (mol _{co} ·g _{Pt} ⁻¹ ·h ⁻¹)	TOF (s ⁻¹)	Reference
0.3Pt/(TiO ₂ -T-500)	62.7	6.15	This work
0.3Pt/(TiO ₂ -A)	31.4	3.54	This work
Pt/TiO ₂	7.42	0.46	Panagiotopoulou, P. et al. <i>J. Catal.</i> 2009 , 267, 57–66.
Pt/0.06Na/TiO ₂	27.6	1.58	Zhu, X. et al. <i>J. Catal.</i> 2011 , 278, 123-132.
Pt-4Na/TiO ₂	7.74	1.39	Chen, Y. et al. <i>ACS Catal.</i> 2018 , 8, 859–868.
Pt/FeO _x	17.2	0.93	Panagiotopoulou, P. et al. <i>Catal. Today</i> 2007 , 127, 319-329.
Pt/Al ₂ O ₃	0.35	0.02	Pierre, D. et al. <i>Top. Catal.</i> 2007 , 46, 363-373.
Pt/CeO _x	1.71	0.17	Zhai, Y. et al. <i>Science</i> 2010 , 329, 1633-1636.
Pt=Na/SiO ₂	4.32	0.24	

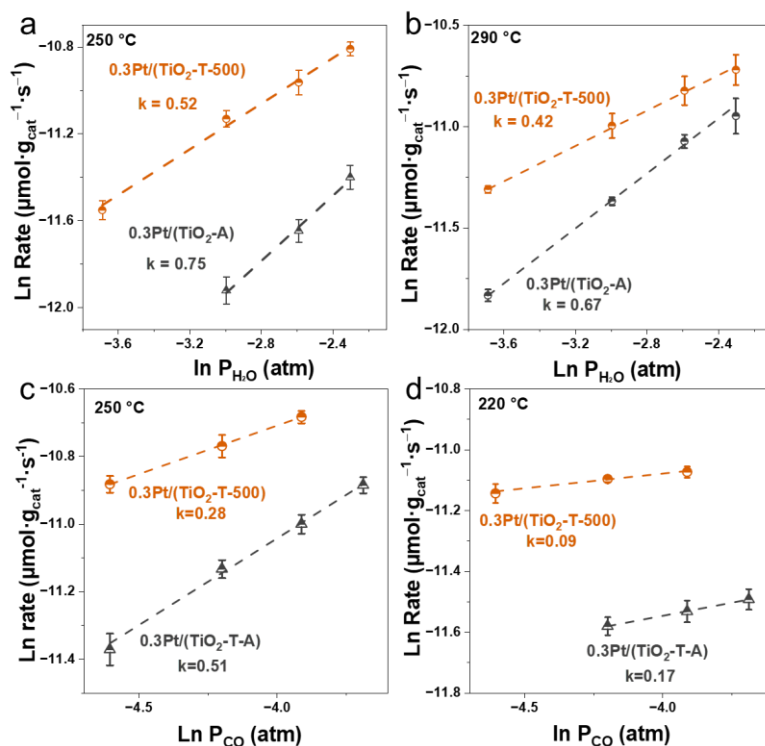


Figure R1-1. (a–b) H_2O reaction orders of $0.3\text{Pt}/(\text{TiO}_2\text{-T-500})$ and $0.3\text{Pt}/(\text{TiO}_2\text{-T-A})$ at (a) 250°C and (b) 290°C . (c–d) CO reaction orders of $0.3\text{Pt}/(\text{TiO}_2\text{-T-500})$ and $0.3\text{Pt}/(\text{TiO}_2\text{-T-A})$ at (c) 250°C and (d) 220°C .

Comment 15: "hydroxyl content on the support to obtain the target Pt/TiO₂ catalyst was shown in Figure 1a."

Grammar: "is shown"

Response: We sincerely appreciate the reviewer's careful review and apologize for the mistake. The manuscript has been revised accordingly on Page 5, line 3.

2. Language and Clarity

The introduction is relatively well-written, but the remainder of the manuscript suffers from frequent grammatical errors and awkward phrasing. Just 2 quick examples:

- *"Commercial anatase (denoted as TiO₂-A) as precursor was hydroxylated by... which denoted as".*

- *"The surface structure of modified TiO₂ and its effects on the state of supported Pt species need to be clearly characterized".*

Response: We are grateful for the reviewer's meticulous and expert suggestions on the wording and expression. The two specific examples of language issues highlighted by the reviewer have been addressed as follows: (1) "**Commercial anatase (denoted as TiO₂-A) was hydrothermally treated with the highly concentrated NaOH solution (10 mol/L), followed by washing with HCl solution.**" and (2) "**It was imperative to elucidate the surface structure of the modified TiO₂ and the chemical state of the supported Pt species.**" on Page 5, lines 4–8. Furthermore, we have performed the comprehensive revision throughout the revised manuscript.

3. Comments about Experimental Issues and Data Interpretation

Comment 1: FTIR Spectra: The materials appear extremely wet in the FTIR results in Fig S2. All samples should be dried (preferably >100 °C) prior to spectral acquisition to allow proper comparison of hydroxyl features and FTIR spectra for all samples should be shown.

Response: Thanks for reviewer's valuable suggestions. The drying temperature for the TiO₂-T-UC support has been increased from the initial 80 °C to 120 °C and maintained for 18 hours. As shown in Figure R1-2a, the TGA results indicated that the mass loss below 200 °C was reduced, suggesting that the increased drying temperature facilitated the removal of physically adsorbed H₂O. Above 200 °C, the mass loss showed no significant change, demonstrating that the surface hydroxyl groups are preserved. FT-IR analysis revealed the slight decrease in the OH signal after drying at 120 °C, which remained significantly stronger than that of the TiO₂-A support (Figure R1-2b). Subsequently, all supports were subjected to FT-IR characterization (Figure R1-2c). It

was observed that the intensity of the OH signal gradually decreased with increasing calcination temperature of the supports, further demonstrating that the calcination temperature plays a regulatory role in the surface hydroxyl content.

We have updated the data for TiO₂-T-UC dried at 120 °C from Figures R1-2a and b to Figure S2 in the supplementary information. The corresponding description in the method section has been revised to "dried in air at 120 °C for 18 hours" on Page 17, line 26 in the revised manuscript (highlight in yellow). Moreover, the Figure R1-2c has been shown as Figure S9b in the supplementary information.

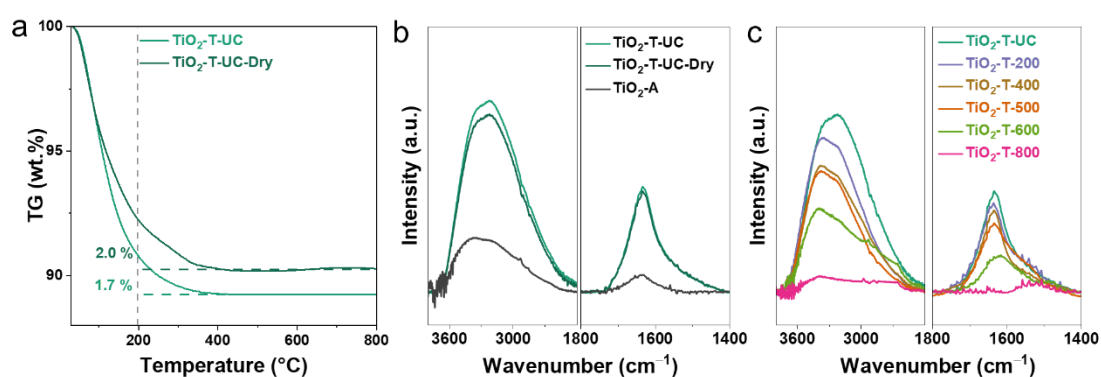


Figure R1-2. (a) TGA curves of TiO₂-T-UC at 80 °C and 120 °C. (b) The attenuated total reflectance-infrared spectroscopy (ATR-IR) of TiO₂-T-UC at 80 °C and 120 °C and TiO₂-A. (c) The attenuated total reflectance-infrared spectroscopy (ATR-IR) of TiO₂-T calcined at various temperatures.

Comment 2: Claims of Pt Encapsulation

(1) The conclusion that Pt is "embedded" in the TiO₂ lattice is based on a single triangular particle image (Supplementary Figs. S3, S4). This is insufficient. Support this claim with catalytic activity suppression or CO chemisorption data to show surface-area loss.

Response: Thanks to the reviewer for the helpful suggestions. The reaction rate of 0.3Pt/(TiO₂-T-UC) was measured to be 4.9 $\mu\text{mol}\cdot\text{g}_{\text{cat}}^{-1}\cdot\text{s}^{-1}$ at 250 °C, which represented the significant decrease compared to the 0.3Pt/(TiO₂-A) catalyst with the non-hydroxylated support (Figure R1-3). When the TiO₂-T-UC support was calcined at 500 °C followed by subsequent Pt loading, superior catalytic activity was achieved. This result corresponded to the HAADF-STEM observations, indicating that the encapsulation of Pt particles led to the suppression of the catalytic activity. Additionally, CO pulse chemisorption experiments were performed on the spent catalysts (Table R1-2). The results revealed that the 0.3Pt/(TiO₂-A) and 0.3Pt/(TiO₂-T-500) catalysts exhibited comparable CO uptake, which was significantly greater than that of the 0.3Pt/(TiO₂-T-UC) catalyst. This indicated that the Pt species in 0.3Pt/(TiO₂-T-UC) were covered, thereby losing their CO adsorption capacity. This conclusion was consistent with the trends observed in catalytic activity, HAADF-STEM images and *in situ* DRIFTS results. **The Figure R1-3 was shown as Figure S5 in the supplementary information. The Table R1-2 was shown as Table S1 in the supplementary information. The corresponding description has been added in the revised manuscript on Page 5, lines 17–19 (highlight in yellow).**

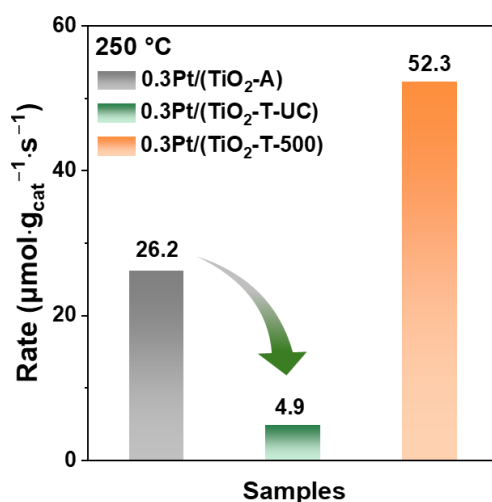


Figure R1-3. The reaction rate of 0.3Pt/(TiO₂-A), 0.3Pt/(TiO₂-T-UC) and 0.3Pt/(TiO₂-T-500) catalysts at 250 °C.

Table R1-2. The CO uptake and Pt dispersion of 0.3Pt/(TiO₂-A), 0.3Pt/(TiO₂-T-UC) and 0.3Pt/(TiO₂-T-500) catalysts.

	CO uptake ($\mu\text{mol/g}$)	Dispersion
0.3Pt/(TiO ₂ -A)	7.4	48.1 %
0.3Pt/(TiO ₂ -T-UC)	2.9	18.8 %
0.3Pt/(TiO ₂ -T-500)	8.5	55.2 %

(2) Figure S4 only presents a narrow IR range. No comparison with other samples is shown. The term "used" is ambiguous; please clarify whether this means spent catalyst.

Response: We thank the reviewer for the valuable suggestions. The CO adsorption results from *in situ* DRIFTS spectra for the three samples after the WGS reaction are presented in Figure R1-4. The results showed that the 0.3Pt/(TiO₂-A) catalyst exhibited distinct signals corresponding to Pt²⁺ and Pt ^{δ +} species. In contrast, the CO adsorption signals for Pt⁰ species were nearly absent in the 0.3Pt/(TiO₂-T-UC) catalyst, which was attributed to the encapsulation of Pt species by the support. Furthermore, upon the adjustment of the hydroxyl group content on the support, pronounced signals of Pt⁰ species re-emerge in the 0.3Pt/(TiO₂-T-500) catalyst, which demonstrated the exposure of the Pt species. The comparison among the three samples provided further evidence for the encapsulation of Pt species by the support in the 0.3Pt/(TiO₂-T-UC) catalyst. **The Figure R1-4 was shown as Figure S4 in the supplementary information and corresponding description has been added in the revised manuscript on Page 5, lines 17–19 (highlight in yellow).**

In addition, the term "used" in this manuscript means spent catalyst, which was indeed ambiguous. Therefore, in accordance with the suggestion of the reviewer, we

have replaced "used" with "spent" throughout the revised manuscript. We thank the reviewer again for the expert comment.

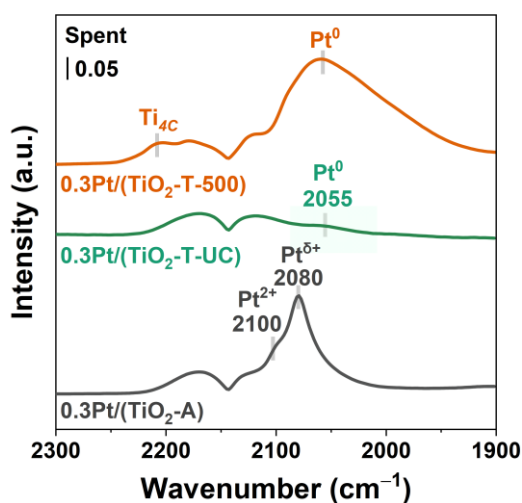


Figure R1-4. *In situ* DRIFTS spectra of the CO adsorption for the spent 0.3Pt/(TiO₂-A), 0.3Pt/(TiO₂-T-UC) and 0.3Pt/(TiO₂-T-500) samples.

Comment 3: Figure 1d – Volcano Plot: 4 data points are insufficient to define a volcano curve. Error bars are missing (everywhere in the article; please add). Lines should be described in the caption and justified physically.

Response: We thank the reviewer for the insightful comments. **The reaction rates of catalysts with supports calcined at 400 °C and 600 °C have been added to Figure R1-5.** It was observed that the Pt-based catalyst supported on TiO₂-T calcined at 500 °C exhibited the optimal catalytic activity. **However, the divergent catalytic activities exhibited by the Pt catalysts supported on TiO₂-T calcined at different temperatures were influenced by the combination of factors.** Firstly, the surface hydroxyl content progressively decreased with increasing calcination temperature, which was confirmed by TGA and ATR-IR results (Figure R1-6). Furthermore, the exposure of Pt species and their particle size also varied with the calcination temperature of the support. As shown in Figure R1-7a, the catalysts with supports that were untreated retained their nanotubular morphology after the WGS reaction.

Interestingly, the Pt species in these samples were distributed both on the inner and outer walls of the nanotubes, resulting in the non-uniform size distribution. In addition to the inadequate exposure of active sites for Pt particles located within the nanotube interior, those on the external surfaces were also found to be embedded within the tube walls. These masked sites collectively contribute to the suppression of WGS activity. When the support was calcined at 200 °C, the majority of the nanotubes were retained (Figure R1-7b). Similarly to the 0.3Pt/(TiO₂-T-UC) catalyst, Pt particles were incorporated into both the interior and the outer walls of these nanotubes, which hindered the exposure of active sites. In contrast to 0.3Pt/(TiO₂-T-UC), a portion of the nanotubes in the 0.3Pt/(TiO₂-T-200) catalyst collapsed, forming layered sheet-like structures. Although the Pt particles supported on these structures still exhibited a non-uniform size distribution, this morphological change increased the number of exposed Pt sites to absorb the CO molecules (Figure R1-8), thereby leading to a slight enhancement in the WGS activity. As shown in Figure R1-9, when the calcination temperature of the support was increased to 400 °C, the TiO₂ nanotubes underwent complete collapse, transforming into the sheet-like morphology. This transformation was accompanied by a progressive increase in the size of the supported Pt particles with rising calcination temperature of TiO₂ support, which was attributed to the loss of surface hydroxyl groups and the consequent weakening of their anchoring effect on the Pt species. **Therefore, Figure R1-5, which showed the variation of catalyst reaction rate with support calcination temperature, served to identify the optimal calcination temperature of 500 °C. Since the volcano-shaped trend lacks physical significance, we have removed the line. The Figure R1-5 has been shown as Figure 1c in the revised manuscript. The Figure R1-6, R1-7, R1-8 and R1-9 has been shown as Figure S9, S7, S8 and S10 in the supplementary information, respectively. The corresponding description has been added on Page 5, lines 20–27 (highlight in yellow).**

In addition, error bars have been incorporated into the reported values for the reaction rates (Figure R1-5), reaction orders (Figure R1-10), and *E_a* (Figure R1-11).

The Figure R1-10a has been shown as Figure 3f in the revised manuscript. The Figure R1-10b has been shown as Figure S31 in supplementary information. The Figure R1-10c has been shown as Figure 2g in the revised manuscript. The Figure R1-10d has been shown as Figure S20 in supplementary information. Figure R1-11 has been shown as Figure S34 in supplementary information.

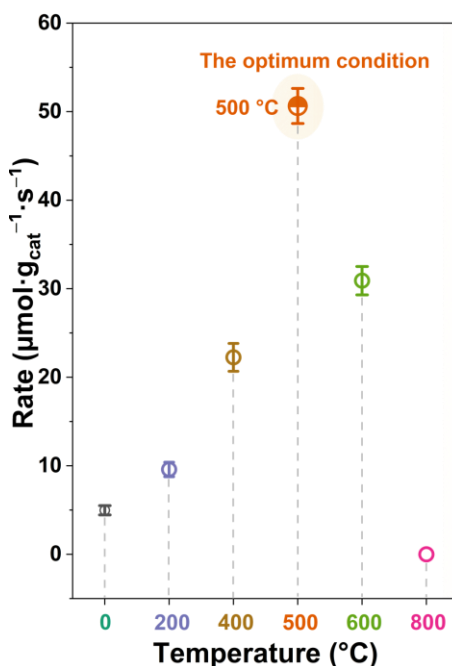


Figure R1-5. WGS reaction rate at 250 °C as the function of calcination temperature of TiO₂-T support, with the optimum at 500 °C.

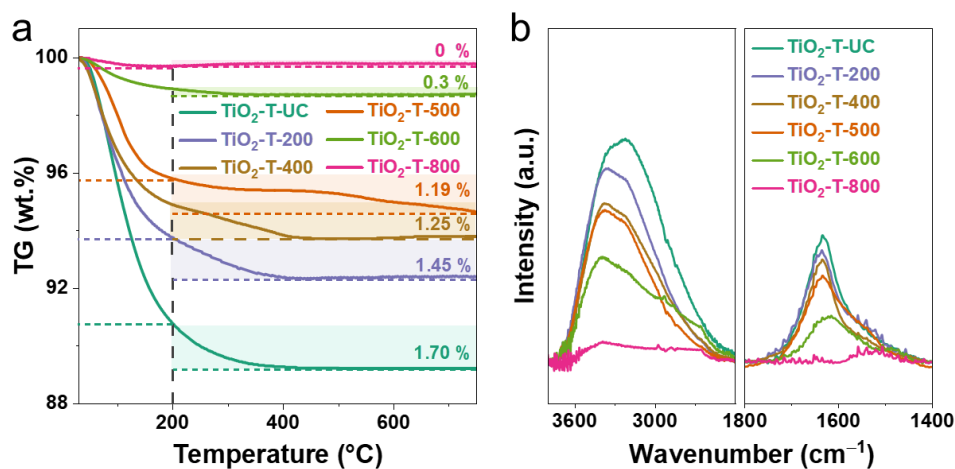


Figure R1-6. (a) TGA curves and (b) The ATR-IR spectra of various TiO_2 under different thermal treatment temperature.

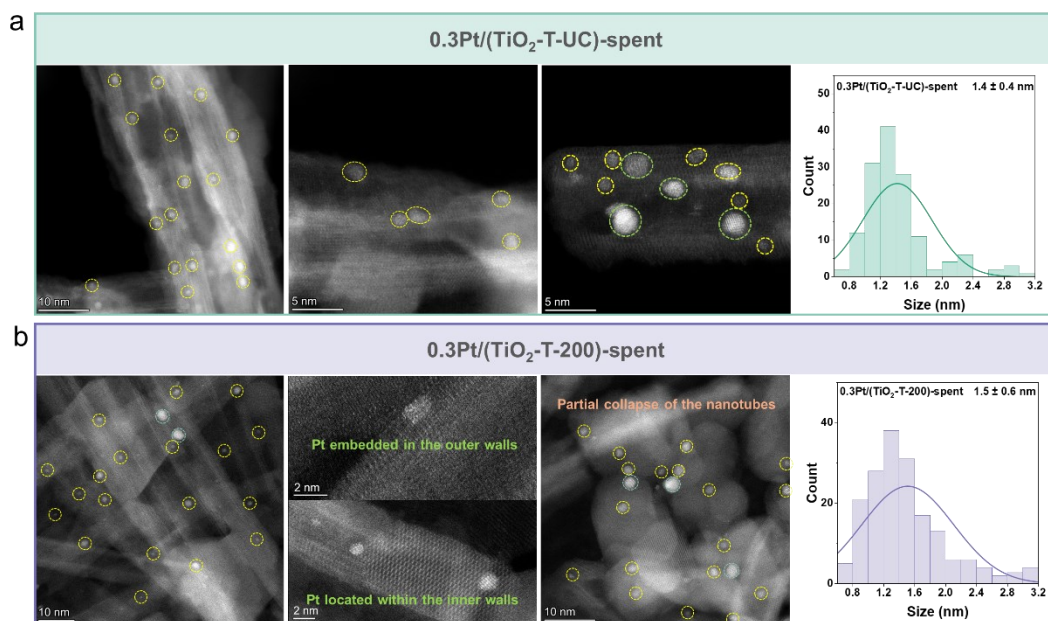


Figure R1-7. The aberration-corrected HAADF-STEM images and the size statistics of the spent (a) 0.3Pt/(TiO_2 -T-UC) and (b) 0.3Pt/(TiO_2 -T-200) catalyst.

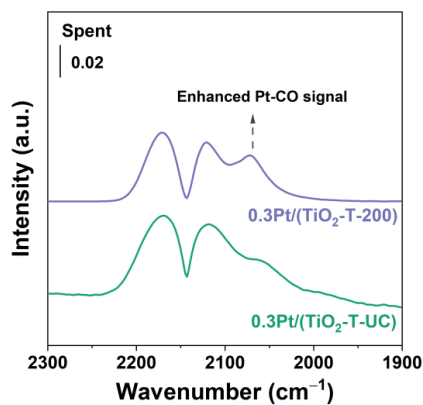


Figure R1-8. *In situ* DRIFTS spectra of the CO adsorption for the spent 0.3Pt/(TiO_2 -T-UC) and 0.3Pt/(TiO_2 -T-200) catalyst.

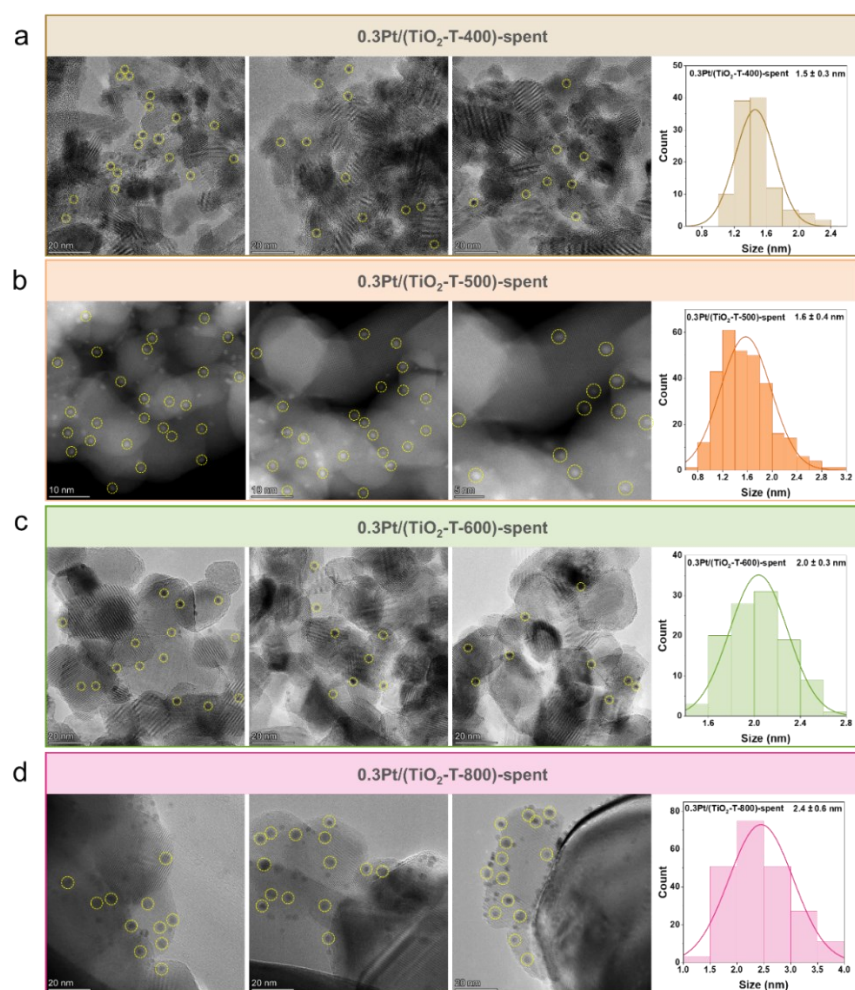


Figure R1-9. The aberration-corrected HAADF-STEM images and the size statistics of the spent (a) 0.3Pt/(TiO₂-T-400), (b) 0.3Pt/(TiO₂-T-500), (c) 0.3Pt/(TiO₂-T-600) and (d) 0.3Pt/(TiO₂-T-800) catalyst.

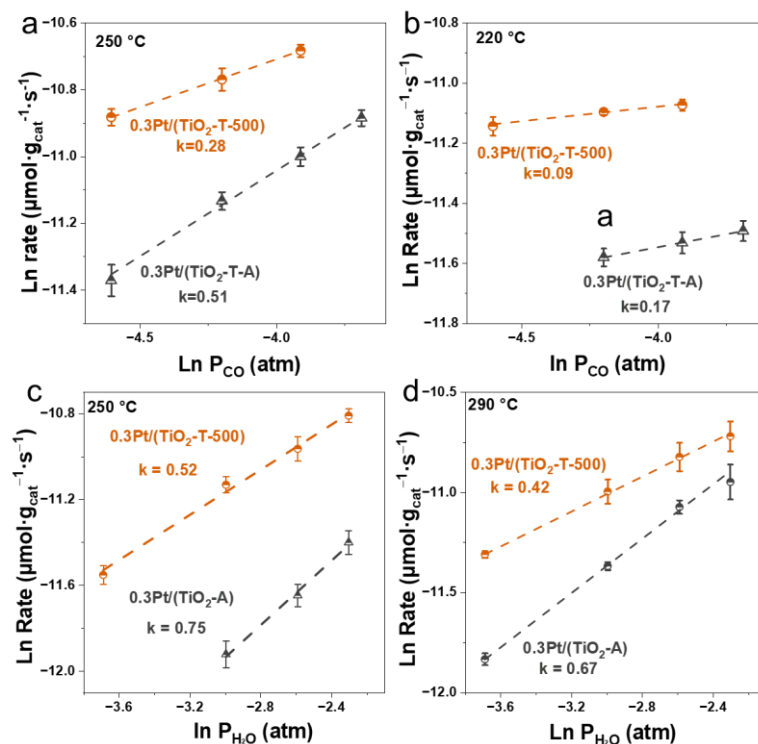


Figure R1-10. The CO reaction order of 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) at (a) 250 °C and (b) 220 °C. The H₂O reaction order of 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) at (a) 250 °C and (b) 290 °C.

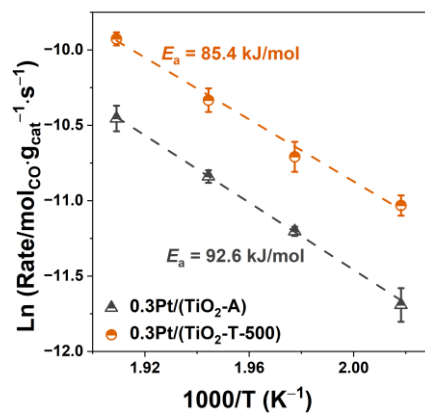


Figure R1-11. Activation energy (E_a) for 0.3Pt/(TiO₂-A) and 0.3Pt/(TiO₂-T-500) catalysts.

Comment 4: Redundancy and Missing Information:

(1) Figure 1f is redundant with 1d.

Response: We thank the reviewer for the thoughtful suggestion. The original Figure 1f has been replaced with a plot of the reaction rate as a function of the support calcination temperature, **which now appeared as Figure 1c in the revised manuscript.** Furthermore, the original Figure 1d revealed that the hydroxylated TiO₂-T calcined at 500 °C, which exhibited optimal catalytic performance after supporting Pt species, retained the higher concentration of surface hydroxyl groups compared to the anatase TiO₂-A precursor. This demonstrated that the two-step hydroxylation–thermal treatment process applied to TiO₂-A successfully produced the TiO₂-T-500 support with the optimal hydroxyl content, thereby resulting in superior WGS activity after Pt loading. **The original Figure 1d has been shown as Figure 1e in the revised manuscript.**

(2) Figure 1e should show TGA of samples treated at 200 and 800 °C.

Response: We thank the reviewer for the careful comments. The surface hydroxyl content of the supports calcined at different temperatures was measured by TGA. It was found that the mass loss above 200 °C gradually decreased with increasing calcination temperature, indicating the reduction in surface hydroxyl groups. Therefore, the TGA results for the 200 °C and 800 °C samples, along with those of other calcination temperatures, are summarized in Figure R1-12. After identifying 500 °C as the optimal support calcination temperature based on the highest catalytic activity, the hydroxyl contents on TiO₂-T-500 and the precursor anatase TiO₂-A were compared (Figure R1-13). The results demonstrated that **a certain amount of hydroxyl groups remained on the TiO₂-T-500 support, and this amount was higher than that on the TiO₂-A support.** Figure R1-12 has been shown as Figure S9a in supplementary information. Figure R1-13a and b has been shown as Figure 1d and e in the revised

manuscript. The corresponding description has been updated on Page 5, lines 23, 24 and Page 7, lines 1–3 in the revised manuscript (highlight in yellow).

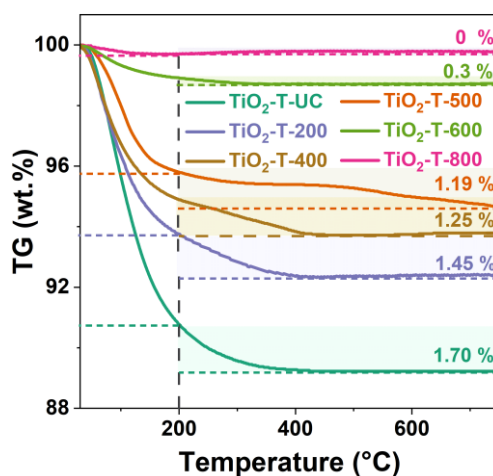


Figure R1-12. TGA curves of various TiO_2 under different thermal treatment temperature.

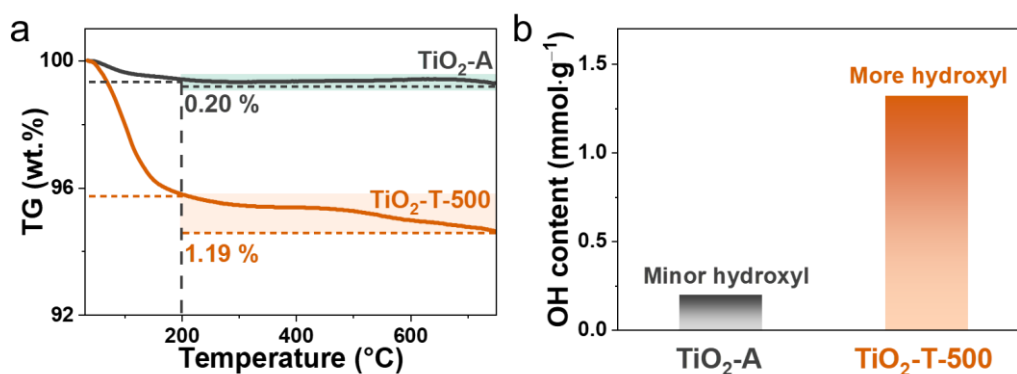


Figure R1-13. (a) TGA curves of $\text{TiO}_2\text{-A}$ and $\text{TiO}_2\text{-T-500}$ supports. (b) The hydroxyl content of $\text{TiO}_2\text{-A}$ and $\text{TiO}_2\text{-T-500}$ supports determined from TGA measurements.

(3) Figure S7 does not convincingly show a significant difference in OH groups. The conclusion of "significantly more abundant" is not justified.

Response: We thank the reviewer for the timely comments. The peak area of the HD signal in the H–D exchange experiment was representative of the surface hydroxyl group concentration. As shown in Figure R1-14b, $\text{TiO}_2\text{-T-500}$ exhibited the larger HD

signal peak area compared to TiO₂-A, indicating the higher abundance of surface hydroxyl groups on TiO₂-T-500. As correctly noted by the reviewer, the difference in the integrated H-D exchange signal intensities between the two supports should not be described as "significantly". Accordingly, the statement in the revised manuscript has been changed to: **"the hydroxyl groups on TiO₂-T-500 were more abundant than those on TiO₂-A."** To further compare the hydroxyl group concentration on the two supports, ATR-IR results demonstrated that TiO₂-T-500 exhibited the more intense hydroxyl signal compared to TiO₂-A (Figure R1-14c). **Figure R1-14 has been shown as Figure S11 in supplementary information and the corresponding description has been added on Page 7, lines 3–5 in the revised manuscript (highlight in yellow).**

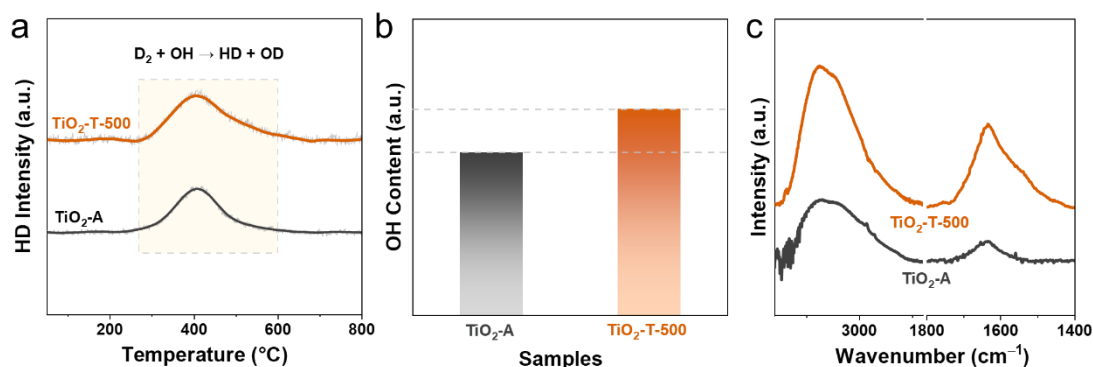


Figure R1-14. (a) HD evolution profiles during the H–D exchange of TiO₂-A and TiO₂-T-500. (b) OH contents of samples determined from the peak area of H–D curves normalized by support mass. (c) ATR-IR of TiO₂-A and TiO₂-T-500.

Comment 5: Particle Size and Dispersion:

(1) No average particle size data are provided for any sample in Figure 1. If particle size changes, so will activity, regardless of hydroxyl concentration.

Response: We thank the reviewer for the expert comments. For the hydroxylated TiO₂-T supports, in the cases of 0.3Pt/(TiO₂-T-UC) and 0.3Pt/(TiO₂-T-200), the preserved nanotubular morphology led to the coverage of Pt active sites, resulting in lower

catalytic activity (Figure R1-7 and R1-8). When the calcination temperature of the support exceeds 400 °C, the nanotubes completely collapse into the sheet-like morphology, leading to full exposure of the Pt sites. However, with the further increase in the calcination temperature, the decreasing hydroxyl group concentration weakened the anchoring ability for the Pt species, leading to the gradual increase in Pt particle size (Figure R1-9). **Thus, the observed variation in catalytic activity of the Pt-based catalysts with support calcination temperature results from the collective influence of multiple factors, including surface hydroxyl content, the exposure of Pt species, and particle size. Consequently, the Figure R1-5 displaying the catalyst reaction rate as a function of support calcination temperature served to identify the optimal thermal treatment conditions for the hydroxylated support.**

The 0.3Pt/(TiO₂-T-500) catalyst, with the support calcined at 500 °C, exhibited the optimal catalytic performance and was therefore selected as the primary subject for comparison with 0.3Pt/(TiO₂-A), which used non-hydroxylated anatase TiO₂ as the support. As shown in Figure R1-15–R1-17, HAADF-STEM characterization revealed that in the fresh 0.3Pt/(TiO₂-T-500) catalyst, Pt species primarily coexisted as single atoms and clusters, which was consistent with the state of Pt species in the fresh 0.3Pt/(TiO₂-A) catalyst. After the WGS reaction, aggregation of Pt species occurred in both catalysts. Statistical analysis of Pt species indicated that the average Pt particle size in the spent 0.3Pt/(TiO₂-T-500) catalyst was 1.6 nm, which was quite similar to the 1.7 nm observed in the spent 0.3Pt/(TiO₂-A) catalyst (Figure R1-18). Moreover, CO chemisorption measurements revealed that the dispersions of the spent 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) catalysts were 48.1 % and 55.2 %, respectively. This result demonstrated that the Pt species in both catalysts exhibited the similar dispersion (Table R1-2). Given the comparable Pt particle size and dispersion in the 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) catalysts, the difference in their catalytic performance was primarily attributed to the distinct concentration of surface hydroxyl groups on the respective supports, which critically influences both the electronic structure of the Pt species and the surface properties of the support. **Figure R1-15 has been shown as Figure 1f–k in**

the revised manuscript. Figure R1-16– R1-18 has been shown as Figure S12–14 in supplementary information. Table R1-2 has been shown as Table S1 in supplementary information. The corresponding description has been revised on Page 5, lines 25–27 and Page 7, lines 9–19 in the revised manuscript (highlight in yellow).

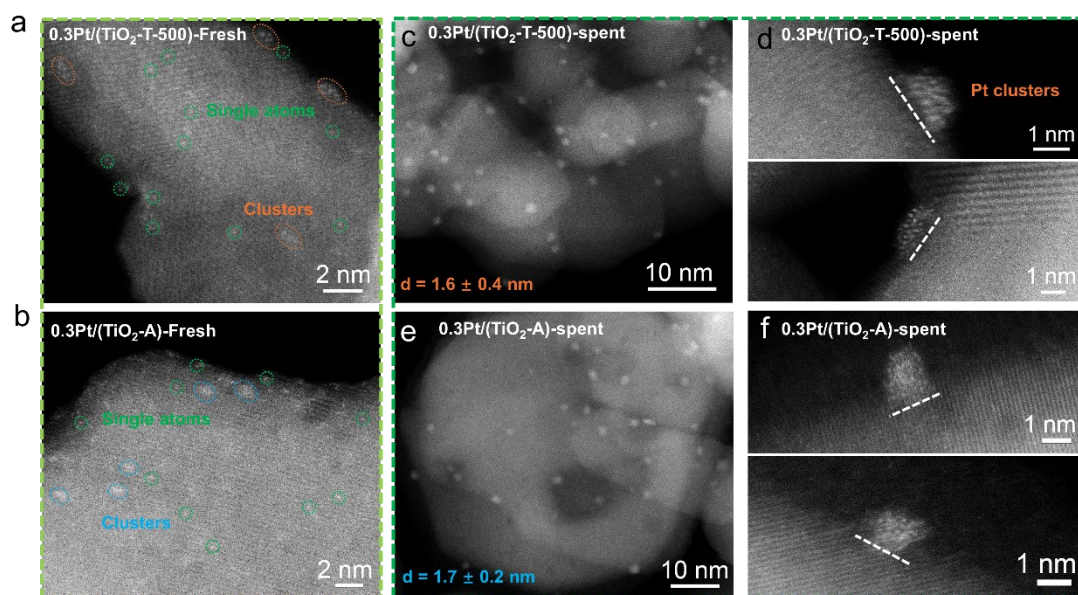


Figure R1-15. (a–f) Aberration-corrected HAADF-STEM images of the fresh (a) 0.3Pt/(TiO₂-T-500) and (b) 0.3Pt/(TiO₂-A) catalyst. (c–f) Aberration-corrected HAADF-STEM images of the spent (c, d) 0.3Pt/(TiO₂-T-500) and (e, f) 0.3Pt/(TiO₂-A) catalyst.

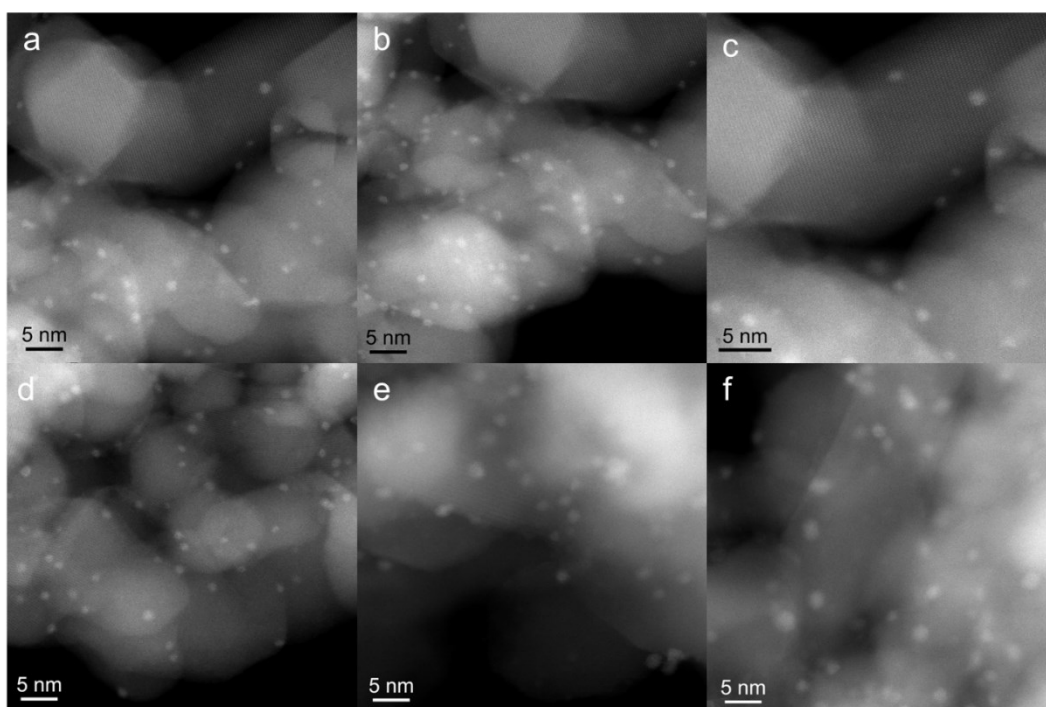


Figure R1-16. (a–f) Aberration-corrected HAADF-STEM images of the spent 0.3Pt/(TiO₂-T-500).

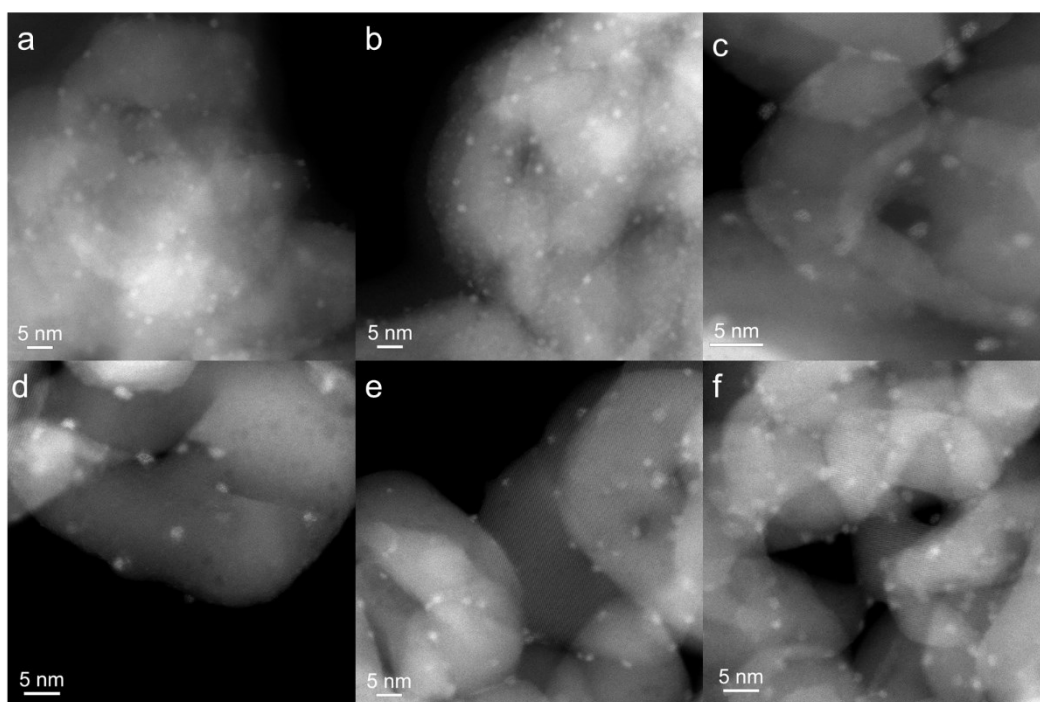


Figure R1-17. (a–f) Aberration-corrected HAADF-STEM images of the spent 0.3Pt/(TiO₂-A).

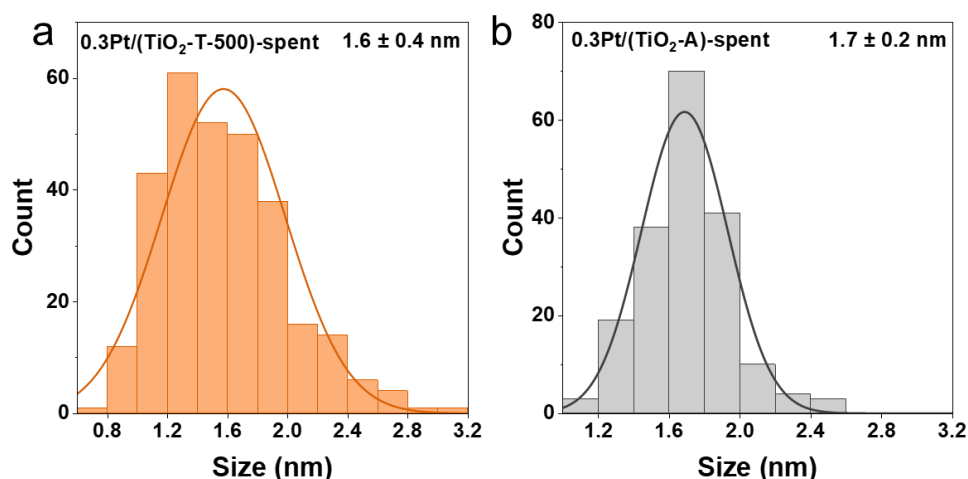


Figure R1-18. The size statistics of the spent (a) 0.3Pt/(TiO₂-T-500) and (b) 0.3Pt/(TiO₂-A) catalysts.

Table R1-2. The CO uptake and Pt dispersion of 0.3Pt/(TiO₂-A), 0.3Pt/(TiO₂-T-UC) and 0.3Pt/(TiO₂-T-500) catalysts.

	CO uptake ($\mu\text{mol/g}$)	Dispersion
0.3Pt/(TiO ₂ -A)	7.4	48.1 %
0.3Pt/(TiO ₂ -T-UC)	2.9	18.8 %
0.3Pt/(TiO ₂ -T-500)	8.5	55.2 %

(2) Provide HAADF-STEM with statistical analysis (>100 particles), histograms, and support with chemisorption data to report surface exposure.

Response: We appreciate the constructive comments from the reviewer. HAADF-STEM images of the spent 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) catalysts have been provided (Figure R1-15–R1-17), along with particle size statistics (Figure R1-18). The results showed that the average particle sizes of the two catalysts are 1.6 nm and 1.7

nm, respectively. In addition, CO chemisorption experiments conducted after the reaction revealed that the Pt dispersions in the 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) catalysts are 48.1% and 55.2%, respectively (Table R1-2). These findings indicated that the Pt species in both catalysts exhibit similar particle sizes and dispersion.

(3) Report TOF values.

Response: We thank the reviewers for the professional comments. The TOF values of the 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) catalysts were 6.15 s⁻¹ and 3.54 s⁻¹, respectively, indicating that the activity per Pt site in the 0.3Pt/(TiO₂-T-500) catalyst was higher than that in the 0.3Pt/(TiO₂-A) catalyst (Table R1-1). **Table R1-1 has been shown as Table S4 in supplementary information. The corresponding calculation formula has been added into the revised manuscript on Page 20, lines 9–17. The corresponding description has been added on Page 16, lines 14–16 in the revised manuscript (highlight in yellow).**

Comment 6: TPR Analysis:

(1) The H₂-TPR data (Fig. 2a) do not prove hydrogen spillover. Lower reduction temperatures can also arise from water loss or hydroxyl dehydrogenation. FTIR suggests the material is wetter; not necessarily more reducible.

Response: We thank the reviewer for the insightful comments. To exclude the potential influence of H₂O desorption during the heating process on the decrease in the reduction peak temperature, H₂-TPR coupled with mass spectrometry was performed on both catalysts (Figure R1-19a). The results indicated that the hydrogen consumption temperature of the 0.3Pt/(TiO₂-T-500) catalyst remained lower than that of 0.3Pt/(TiO₂-A). Moreover, no significant H₂O signal was detected simultaneously with the H₂ consumption peaks. This observation was likely attributable to the fact that both supports were calcined at 500 °C and subsequently the catalyst pretreated at 120 °C

under the Ar atmosphere prior to the H₂-TPR measurements. **The findings demonstrated that the lower reduction temperature did not arise from the water loss.** Furthermore, the TiO₂-T-500 and TiO₂-A supports exhibited negligible TCD signals during H₂-TPR analysis (Figure R1-19b). This indicated that the surface hydroxyl groups did not consume H₂, **thereby ruling out the possibility that the lower reduction temperature was caused by dehydrogenation of hydroxyl groups.**

Subsequently, we further quantified the H₂ consumption from the H₂-TPR results of the two catalysts (Figure R1-19c). It was found that the actual H₂ consumption of the 0.3Pt/(TiO₂-T-500) catalyst exceeded the theoretical H₂ consumption of the Pt species, whereas the actual H₂ consumption of 0.3Pt/(TiO₂-A) was less than its theoretical Pt consumption. Although this evidence could not directly prove the occurrence of hydrogen spillover, it clearly demonstrated that the Pt species in the 0.3Pt/(TiO₂-T-500) catalyst activated the support, thereby leading to enhanced reducibility.

The Figure R1-19 has been shown as Figure S15 in the supplementary information and the corresponding description has been added on Page 7, lines 27–28 and Page 8, lines 3–4 in the revised manuscript (highlight in yellow).

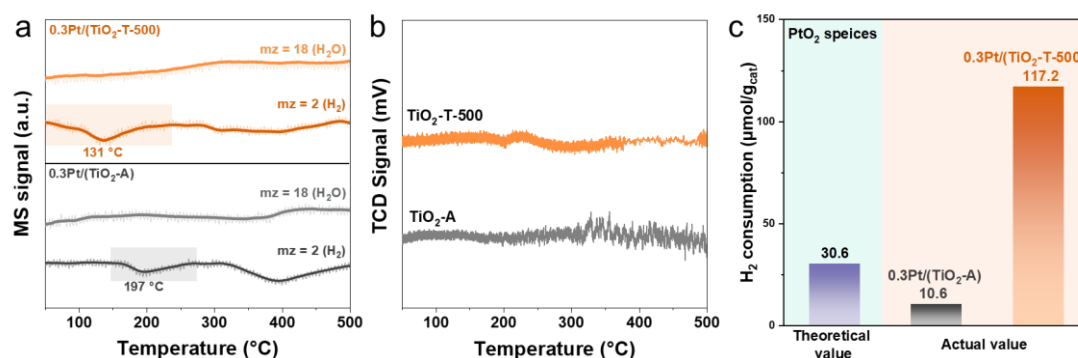


Figure R1-19. (a) H₂-TPR profiles of the 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) samples characterized by mass spectrometry. (b) H₂-TPR profiles of TiO₂-T-500 and TiO₂-A support. (c) H₂ consumption calculated from the results in H₂-TPR profiles of the 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) samples.

(2) Assigning changes in reducibility to Pt species alone (e.g., $Pt^{2+} \rightarrow Pt^0$) without structural or XAS data is speculative.

Response: We are grateful to the reviewer for the constructive comments. To confirm the evolution of Pt species during H_2 reduction, structural characterization was performed. According to the Pt 4f XPS spectra, the Pt species in both the 0.3Pt/(TiO₂-A) and 0.3Pt/(TiO₂-T-500) catalysts exhibited the binding energy of 72.3 eV and 72.5 eV, which was assigned to Pt^{2+} species (Figure R1-20a and b). After H_2 pretreatment, as shown in Figure R1-20c, the Pt in the 0.3Pt/(TiO₂-A) catalyst showed the binding energy of 71.4 eV, corresponding to $Pt^{\delta+}$. In contrast, the binding energy for the 0.3Pt/(TiO₂-T-500) catalyst shifted down by 0.7 eV to 70.8 eV, indicating the presence of Pt^0 species. This result demonstrated the superior reducibility of the 0.3Pt/(TiO₂-T-500) catalyst compared to 0.3Pt/(TiO₂-A), facilitating the reduction of Pt^{2+} to Pt^0 during H_2 treatment.

Moreover, *In situ* XANES profile at Pt L₃ edge under the H_2 pretreatment was further carried out (Figure R1-21). Despite the relatively poor signal-to-noise ratio of the Pt/TiO₂ catalysts due to the low Pt loading (0.3 wt%), both catalysts were observed that white-line intensity was close to that of Pt foil, indicating the reduction of Pt species under the H_2 atmosphere. Furthermore, the white-line intensity of 0.3Pt/(TiO₂-A) was higher than that of 0.3Pt/(TiO₂-T-500). This white-line intensity was proportional to the transition probability of 2p electrons to unoccupied 5d orbitals (Rossi, G. et al. *Phys. Rev. B* **1983**, 27, 5154–5157). The higher white-line intensity corresponded to the greater number of unoccupied 5d states and the lower 5d-electron density, thus indicating the higher oxidation state of Pt species in 0.3Pt/(TiO₂-A).

Figure R1-20a and b has been shown as Figure S16 in the supplementary information. Figure R1-20c has been revised in Fig. 2b in the revised manuscript. Figure R1-21 has been shown as Figure S17 in the supplementary information. The corresponding description has been added on Page 8, lines 5–12 and 17–21 lines in the revised manuscript (highlight in yellow). The corresponding reference was cited as 50 in the revised manuscript.

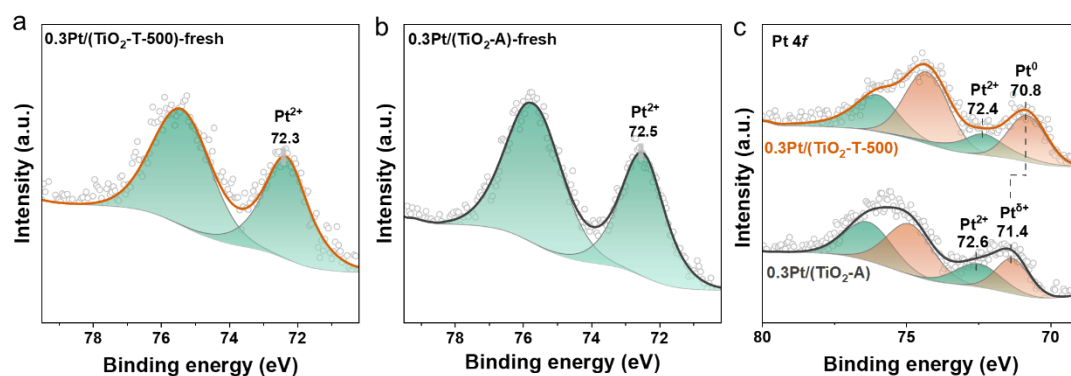


Figure R1-20. (a, b) The Pt 4f XPS for the fresh (a) 0.3Pt/(TiO₂-T-500) and (b) 0.3Pt/(TiO₂-A) samples. (c) The Pt 4f XPS for 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) samples after H₂ pretreatment.

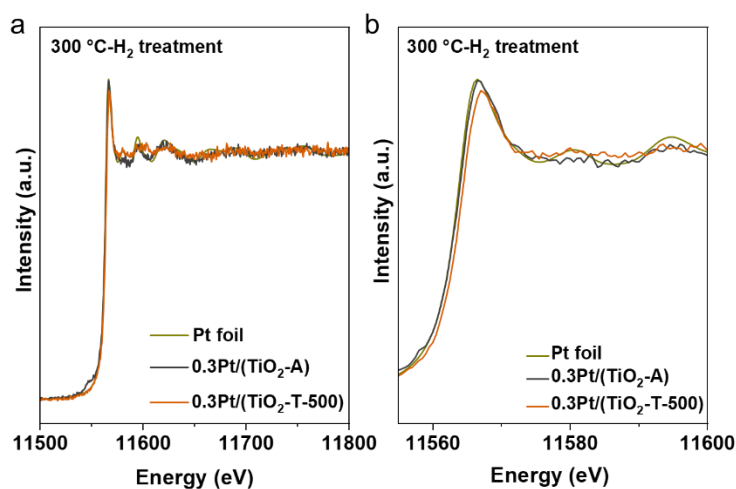


Figure R1-21. *In situ* XANES profile at Pt L₃ edge under the H₂ pretreatment.

Comment 7: *XPS Data:* The XPS spectra (Fig. 1c) show minimal differences. Claims of electron-rich Pt⁰ formation are not distinguishable from noise. Additional XAS data are needed.

Response: We thank the reviewer for the insightful comment. *In situ* XANES profile at Pt L₃ edge under the H₂ pretreatment was further carried out (Figure R1-21). Despite the relatively poor signal-to-noise ratio of the Pt/TiO₂ catalysts due to the low Pt loading (0.3 wt%), the difference in white-line intensity between the two catalysts

under H₂ flow was still distinguishable. **Compared with 0.3Pt/(TiO₂-A), 0.3Pt/(TiO₂-T-500) exhibited the weaker white-line intensity, indicating that Pt species in 0.3Pt/(TiO₂-T-500) possessed higher 5d electron density.** It was noteworthy that the white-line intensity of the Pt species in 0.3Pt/(TiO₂-A) was nearly identical to that of Pt foil, and by comparison, 0.3Pt/(TiO₂-T-500) exhibited the weaker white-line intensity than Pt foil. **We must elucidate the reason for the low white-line intensity of the Pt species occurred in both catalysts.**

(1) As the measurements were performed under the H₂ flow, the Pt L₃-edge white-line intensity decreased and the edge of 0.3Pt/(TiO₂-T-500) shifted to higher energy relative to Pt foil, suggesting the H adsorption on Pt species modulated the Pt electronic state and decreased the unoccupied 5d states (Gorczyca, A. et al. *Angew. Chem. Int. Ed.* **2014**, 53, 12426–12429). Such hydrogen adsorption was also considered to be present in the 0.3Pt/(TiO₂-A) sample. This accounted for the consistently low white line intensity observed in both catalysts. Nevertheless, 0.3Pt/(TiO₂-T-500) exhibited an even lower white line intensity than 0.3Pt/(TiO₂-A), indicating the presence of a greater number of Pt⁰ sites.

(2) XANES results reflect the averaging effect over the bulk structure of the metal species, unlike XPS and DRIFTS spectra, which can distinguish the specific states of different sites in the surface structure of the metal species. Therefore, the XANES results fail to reflect the accurate state of the surface structure of the metal species.

(3) Given the low Pt loading, the catalysts underwent approximately 2 hours of data acquisition during the XANES measurements, during which the Pt species were continuously reduced under the H₂ flow. This was further corroborated by *in situ* DRIFTS spectra after 2 hours of H₂ treatment. The signals at 2060 cm⁻¹ and 2074 cm⁻¹, attributed to Pt⁰ and Pt^{δ+}, respectively, were observed for the 0.3Pt/(TiO₂-A) catalyst (Figure R1-22). These signals might have been averaged in the XANES spectra, resulting in a white-line intensity similar to that of Pt foil. However, since the catalysts were only reduced in H₂ atmosphere for 30 minutes prior to the actual reaction, the

current XANES results did not fully represent the state of the Pt species under real reaction conditions.

Based on the above three points, the Pt species in the two catalysts cannot be unambiguously identified from the XANES results. However, it can be concluded that the Pt sites in the 0.3Pt/(TiO₂-T-500) catalyst possessed the lower oxidation state compared to those in 0.3Pt/(TiO₂-A).

To minimize the interference of noise on the XPS signal, measurements with over 30 scanning cycles were performed (Figure R1-20c). The results consistently revealed the shift in the binding energy of Pt species from 71.4 eV in the 0.3Pt/(TiO₂-A) catalyst to 70.8 eV in the 0.3Pt/(TiO₂-T-500) catalyst, which was attributed as Pt⁰ species, indicating the increase in electron density of Pt species in 0.3Pt/(TiO₂-T-500).

Figure R1-20c has been revised in Figure 2b in the revised manuscript. Figure R1-21 has been shown as Figure S17 in the supplementary information. The corresponding description has been added on Page 8, lines 7–12 and 17–21 in the revised manuscript (highlight in yellow).

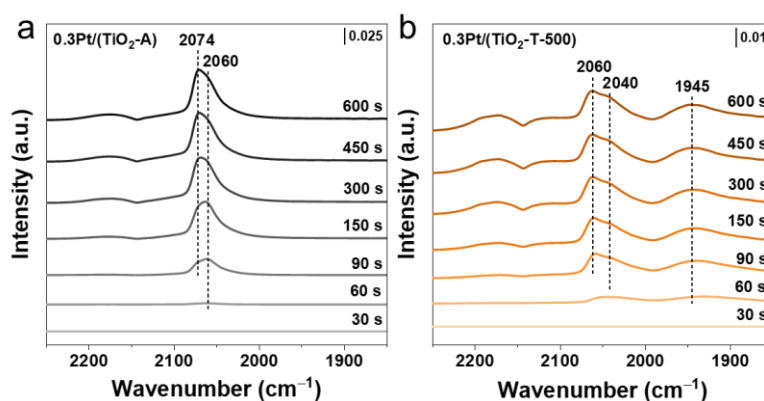


Figure R1-22. *In situ* DRIFTS spectra at 300 °C of (a) 0.3Pt/(TiO₂-A) and (b) 0.3Pt/(TiO₂-T-500) after H₂ pretreatment for 2 hours.

Comment 8: CO Adsorption / DRIFTS Interpretation:

(1) The assignment of bands to Pt⁰ vs. Pt^{δ+} is highly tentative.

Response: We thank the reviewer for the constructive comment. We agree that the assignment of adsorbed species to specific metal sites based solely on *in situ* DRIFTS spectra can be challenging and requires careful justification. We have now strengthened our assignment by providing the more comprehensive discussion based on established literature and well-understood electronic effects.

1. Support from Literature: Our assignment was supported by the literature on CO adsorption on Pt-based catalysts. Many studies using combined techniques such as *in situ* DRIFTS and XAS spectra have reported that the linearly adsorbed CO on Pt⁰ sites typically appears in the region of 2030-2070 cm⁻¹, while CO adsorbed on Pt^{δ+} sites was observed at higher wavenumbers, typically 2070-2095 cm⁻¹. Additionally, the adsorption signal at 1950 cm⁻¹ was attributed to bridge-bonded CO on the interface between metallic Pt clusters and the support (Dessal, C. et al. *ACS Catal.* **2019**, 9, 5752–5759; Zhou, Y. et al. *ACS Catal.* **2016**, 6, 7799–7809; Qiao, B. et al. *Nat. Chem.* **2011**, 3, 634–641; Resasco, J. et al. *J. Am. Chem. Soc.* **2020**, 142, 169–184). Based on the aforementioned literature, the CO adsorption band observed at 2080 cm⁻¹ in the *in situ* DRIFTS of the 0.3Pt/(TiO₂-A) catalyst was assigned to Pt^{δ+} species, whereas the signals at 2064 cm⁻¹, 2043 cm⁻¹, and 1965 cm⁻¹ for the 0.3Pt/(TiO₂-T-500) catalyst were attributed to Pt⁰ clusters.

2. Electronic Effect: The blue shift in the CO adsorption band from Pt⁰ to Pt^{δ+} is attributed to the electronic effects governing CO chemisorption. The Pt^{δ+} site, being electron-deficient, has the reduced capacity for π -back-donation from its d-orbitals to the 2 π^* anti-bonding orbital of the CO molecule. This weakened back-donation results in the stronger C-O bond, which corresponds to a larger force constant (k). According to Hooke's law (eq 1), the force constant k is proportional to the vibrational frequency (ν) and wavenumber ($\tilde{\nu}$) in the corresponding DRIFTS spectra (Zhang, R. et al. *ACS Catal.* **2020**, 10, 14722–14731). Therefore, the higher frequency band we assigned to Pt^{δ+} in the 0.3Pt/(TiO₂-A) catalyst and Pt⁰ in the 0.3Pt/(TiO₂-T-500) was fully consistent with this widely accepted model.

$$\tilde{\nu} = \frac{1}{2\pi c} \sqrt{\frac{k}{\mu}} \quad \text{eq 1}$$

$\tilde{\nu}$ represents the wavenumber; k represents the force constant; μ represents the reduced mass of the vibrational system.

The corresponding description has been added on Page 8, lines 14–17, Page 9, lines 14 and Page 10, lines 1–6. The corresponding reference has been cited as 45–49.

(2) CO-TPD experiments only support stronger binding, not overall "adsorption capability."

Response: We thank the reviewer for their valuable comment. We have revised to “**the enhanced binding of CO on the 0.3Pt/(TiO₂-T-500) catalyst**” in the revised manuscript on Page 10, line 9 (highlight in yellow).

***Comment 9:** "CO adsorbed on the electron-deficient Pt δ^+ species", Again, this assignment rests on uncertain XPS data.*

Response: We thank the reviewer for the valuable comments. *In situ* XANES profile at Pt L₃ edge under the H₂ pretreatment was further carried out (Figure R1-21). Despite the relatively poor signal-to-noise ratio of the Pt/TiO₂ catalysts due to the low Pt loading (0.3 wt%), the difference in white-line intensity between the two catalysts under H₂ flow was still distinguishable. **Compared with 0.3Pt/(TiO₂-T-500), 0.3Pt/(TiO₂-A) exhibited the higher white-line intensity, indicating that Pt species in 0.3Pt/(TiO₂-A) possessed lower 5d electron density.** It was noteworthy that the white-line intensity of the Pt species in 0.3Pt/(TiO₂-A) was nearly identical to that of Pt foil, and by comparison, 0.3Pt/(TiO₂-T-500) exhibited the weaker white-line intensity than Pt foil. **We must elucidate the reason for the low white-line intensity of the Pt species occurred in both catalysts.**

(1) As the measurements were performed under the H₂ flow, the Pt L₃-edge white-line intensity decreased and the edge of 0.3Pt/(TiO₂-T-500) shifted to higher energy relative to Pt foil, suggesting the H adsorption on Pt species modulated the Pt electronic state and decreased the unoccupied 5d states (Gorczyca, A. et al. *Angew. Chem. Int. Ed.* **2014**, *53*, 12426–12429). Such hydrogen adsorption was also considered to be present in the 0.3Pt/(TiO₂-A) sample, thereby reducing the observed white line intensity.

(2) XANES results reflect the averaging effect over the bulk structure of the metal species, unlike XPS and DRIFTS spectra, which can distinguish the specific states of different sites in the surface structure of the metal species. Therefore, the XANES results fail to reflect the accurate state of the surface structure of the metal species.

(3) Given the low Pt loading, the catalysts underwent approximately 2 hours of data acquisition during the XANES measurements, during which the Pt species were continuously reduced under the H₂ flow. This was further corroborated by *in situ* DRIFTS spectra after 2 hours of H₂ treatment. The signals at 2060 cm⁻¹ and 2074 cm⁻¹, attributed to Pt⁰ and Pt^{δ+}, respectively, were observed for the 0.3Pt/(TiO₂-A) catalyst (Figure R1-22). These signals might have been averaged in the XANES spectra, resulting in a white-line intensity similar to that of Pt foil. However, since the catalysts were only reduced in H₂ atmosphere for 30 minutes prior to the actual reaction, the current XANES results did not fully represent the state of the Pt species under real reaction conditions.

Based on the above three points, the Pt species in the two catalysts cannot be unambiguously identified from the XANES results. However, it can be concluded that the Pt sites in the 0.3Pt/(TiO₂-A) catalyst possessed the higher oxidation state compared to those in 0.3Pt/(TiO₂-T-500).

The XPS measurements with over 30 scanning cycles were performed, which effectively improved the signal-to-noise ratio (Figure R1-20c). **The results showed the binding energy of Pt species at 71.4 eV in the 0.3Pt/(TiO₂-A) catalyst and 70.8 eV in the 0.3Pt/(TiO₂-T-500) catalyst, which provided stronger evidence that the**

0.3Pt/(TiO₂-A) catalyst exhibited Pt^{δ+} species, while the 0.3Pt/(TiO₂-T-500) catalyst contains Pt⁰ species.

Comment 10: "In contrast, for the 0.3Pt/(TiO₂-T-500) catalyst... CO₂ products... Fig. 2i", This claim is highly questionable. Figure 2h shows a CO₂ peak for the "A" sample that is larger than that for the "T-500" sample. The conclusion that CO₂ is not formed on the A sample is unsubstantiated. Under Ar, no desorption curves are provided for CO, which complicates interpretation. It is also very plausible that the T-500 sample simply adsorbs CO more strongly due to particle size differences.

Response: We thank the reviewer for the professional comments. To identify the origin of the gaseous CO₂ signal observed during *in situ* DRIFTS measurements, mass spectrometry was employed to monitor the catalytic behavior during temperature-programmed WGS reactions. As shown in Figure R1-23, both catalysts exhibited rapid consumption of reactant CO molecules before the occurrence of the WGS reaction. However, a notable difference was observed. The 0.3Pt/(TiO₂-A) catalyst produced only CO₂ without H₂, indicating that abundant surface oxygen on the catalyst facilitated the reaction with CO at relatively low temperatures to form CO₂. In contrast, the 0.3Pt/(TiO₂-T-500) catalyst generated both CO₂ and H₂, demonstrating that the reaction between OH and CO occurred. Therefore, the gaseous CO₂ signal detected by *in situ* DRIFTS during the CO adsorption for 0.3Pt/(TiO₂-A) originated from CO₂ formed by the reaction between CO and surface oxygen.

To eliminate the influence of surface oxygen on the 0.3Pt/(TiO₂-A) catalyst, multiple CO adsorption and purging experiments were conducted during *in situ* DRIFTS testing. As shown in Figure R1-24, the gaseous CO₂ signal progressively weakened with successive CO adsorption cycles and nearly disappeared after the fourth cycle, indicating that surface oxygen was effectively removed. Therefore, after four CO adsorption cycles, switching experiments involving CO and Ar or Ar+H₂O were performed again. As shown in Figure R1-25, at 250 °C, the CO adsorption peak on Pt^{δ+}

species of 0.3Pt/(TiO₂-A) gradually diminished under Ar purging, and upon re-adsorption of CO, the CO signal at 2080 cm⁻¹ re-emerged. Subsequently, as shown in Figure R1-26a, upon switching to the Ar+H₂O atmosphere, the CO adsorption peak similarly diminished gradually, with the decay rate consistent with that observed under pure Ar purging. Meanwhile, the concurrent decrease in the gaseous CO₂ signal occurred. In contrast, for the 0.3Pt/(TiO₂-T-500) catalyst, the CO signal corresponding to Pt⁰ species decayed significantly faster under the Ar + H₂O atmosphere than under pure Ar, accompanied by the gaseous CO₂ signal that initially increased and subsequently decreased (Figure R1-26b). This indicated that CO adsorbed on Pt⁰ sites reacted with H₂O, generating additional CO₂. As the adsorbed CO on Pt⁰ was progressively consumed, the production of CO₂ gradually diminished and eventually vanished upon complete purging. These results collectively demonstrated the superior ability of Pt⁰ species in activating CO molecules.

In summary, the time-dependent CO desorption curves under the Ar atmosphere are summarized in Figure R1-26a (right). Figure R1-24 has been shown as Figure S21 in the supplementary information. Figure R1-25 has been shown as Figure S22 in the supplementary information. Figure R1-26 has been shown as Figure 2h and 2i in the revised manuscript. The corresponding description has been added on Page 10, lines 22–29 and Page 11, lines 1–5.

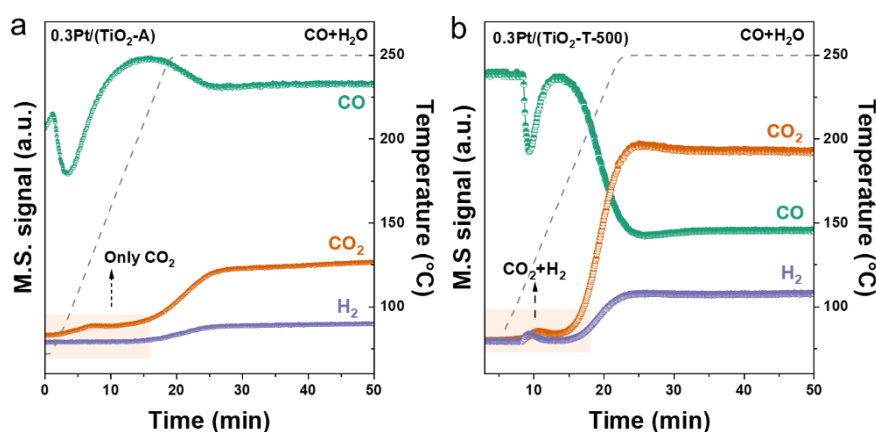


Figure R1-23. The catalytic behavior of (a) 0.3Pt/(TiO₂-A) and (b) 0.3Pt/(TiO₂-T-500) during the heating process of the water gas shift (WGS) reaction.

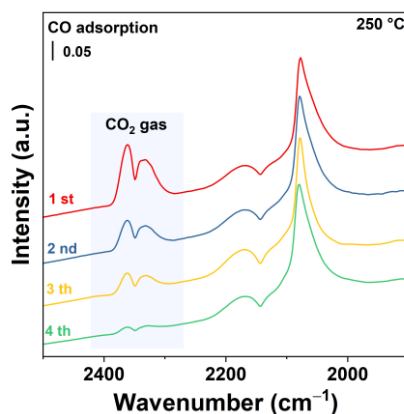


Figure R1-24. Four cycles of CO adsorption experiments conducted under in situ DRIFTS spectroscopy of 0.3Pt/(TiO₂-A) catalyst at 250 °C.

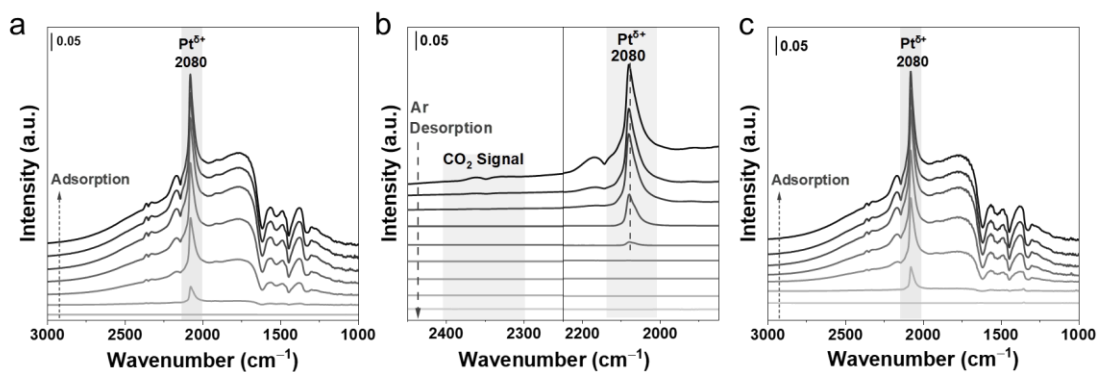


Figure R1-25. *In situ* DRIFTS spectra of (a) CO adsorption, (b) Ar desorption and (c) CO re-adsorption over 0.3Pt/(TiO₂-A) catalyst at 250 °C.

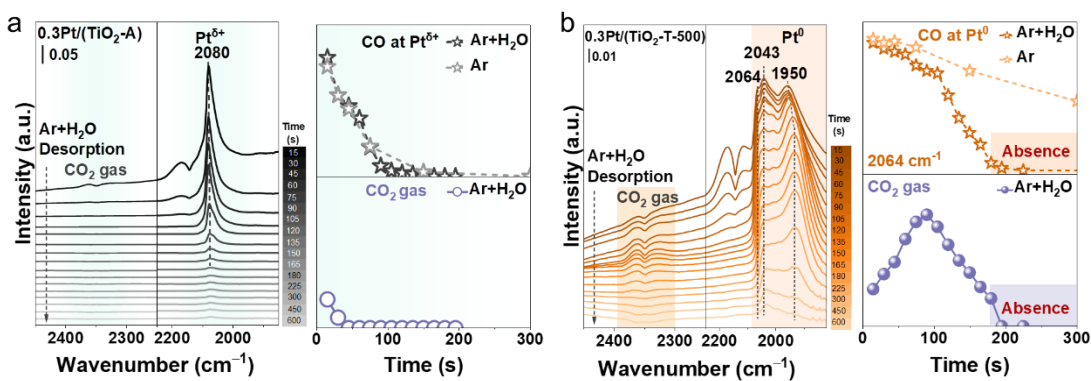


Figure R1-26. (a, b) *In situ* DRIFTS spectra of Ar+H₂O purging over (a) 0.3Pt/(TiO₂-A) and (b) 0.3Pt/(TiO₂-T-500) at 250 °C. Right: the intensity of CO adsorption on Pt sites upon Ar+H₂O and Ar purging as function of time.

Comment 11: Reaction Mechanism Claims: "This corresponded to the conclusion that oxygen vacancies were formed by the consumption of surface hydroxyls by CO during the reaction..." Again, this could also be due to differences in particle size or suboxide shell formation. Particle sizes must be reported for all fresh and spent catalysts.

Response: We thank the reviewer for the perceptive comments. The HAADF-STEM images of the fresh and spent 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) catalysts were presented in Figure R1-27. It was found that Pt species in both fresh catalysts primarily existed as the mixture of single atoms and clusters. After the WGS reaction, these species aggregated to form Pt clusters in both catalysts. According to particle size statistics, the average sizes of the Pt clusters in the spent 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) catalysts were approximately 1.6 nm and 1.7 nm, respectively, which indicated that the Pt clusters in both catalysts exhibited similar dimensions (Figure R1-28). Therefore, the disparity in the concentration of oxygen vacancies between the two spent catalysts originated from the difference in their surface hydroxyl concentrations. These hydroxyl groups were consumed by CO during the reaction, consequently leading to the formation of oxygen vacancies. **The corresponding description has been added on Page 14, lines 13–17 in the revised manuscript (highlight in yellow).**

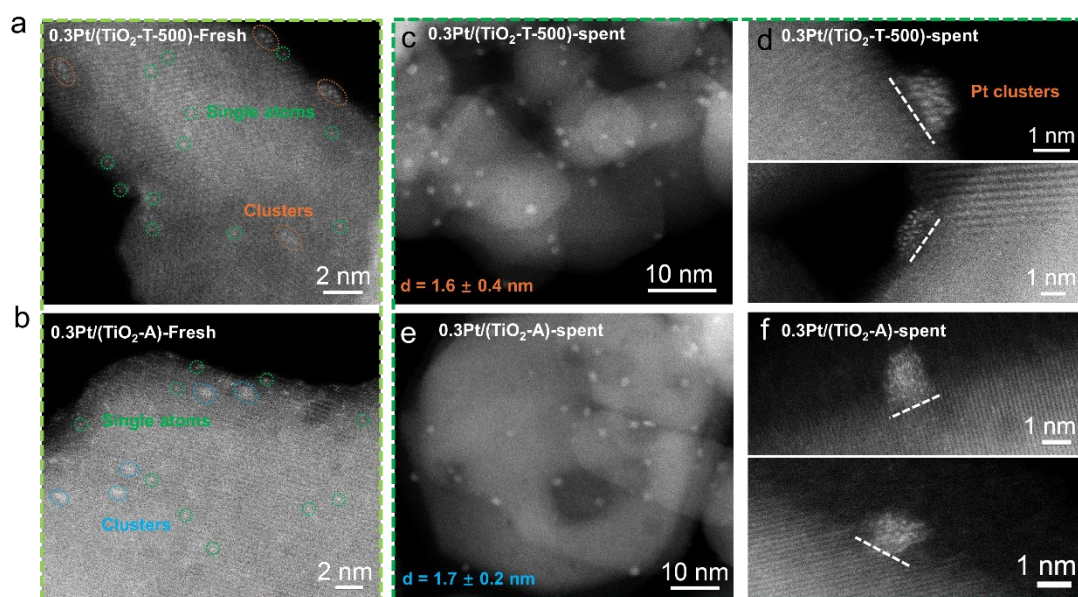


Figure R1-27. (a–f) Aberration-corrected HAADF-STEM images of the fresh (a) 0.3Pt/(TiO₂-T-500) and (b) 0.3Pt/(TiO₂-A) catalyst. (c–f) Aberration-corrected HAADF-STEM images of the spent (c, d) 0.3Pt/(TiO₂-T-500) and (e, f) 0.3Pt/(TiO₂-A) catalyst.

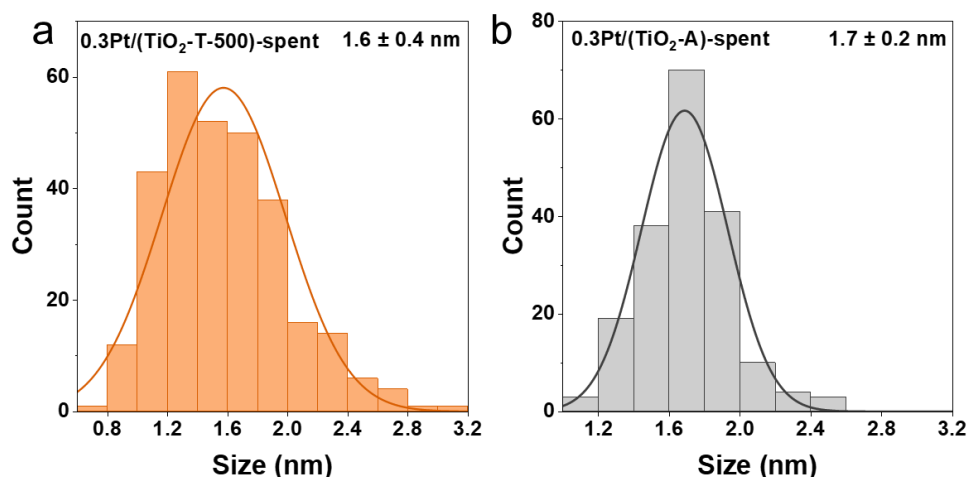


Figure R1-28. The size statistics of the spent 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) catalysts.

***Comment 12:** Overall. This study explores a timely and important topic, but currently suffers from overinterpretation of spectroscopic data, lack of rigorous controls, and incomplete physical characterization. Stronger claims about site-specific activity or electronic structure require corroboration by XAS, chemisorption, and particle size analysis. Many conclusions are not fully supported by the presented data. Substantial revisions are necessary before this manuscript can again be considered for publication in Nature Communications.*

Response: We thank the reviewer for the valuable comments. We have conducted the additional experiments as suggested, including X-ray absorption spectroscopy (XAS), CO chemisorption, particle size analysis and so on. Current results demonstrated that, **with the size effect ruled out, the presence of surface hydroxyl groups on the support modulated the electronic structure of the supported Pt species, leading to**

the formation of 2 nm Pt⁰ clusters, which promoted the adsorption and activation of CO molecules. The description of the spectral results has been revised accordingly. We believe these revisions have significantly improved the quality of the manuscript and hope that the manuscript now can be considered for publication in *Nature Communications*.

Reviewer #2: *Jia and coauthors present a compelling strategy for enhancing the catalytic performance of supported metal-oxide catalysts by modulating the electronic structure of active sites through surface hydroxyl groups. This work convincingly demonstrates the formation of electron-rich Pt^0 clusters (~ 2 nm) on hydroxylated TiO_2 , in contrast to conventional $Pt^{\delta+}$ -Ov- Ti^{3+} interfacial structures. This modification not only enhances CO adsorption and activation but also couples with hydroxyl-assisted oxygen vacancy formation to promote H_2O dissociation, leading to synergistic improvement in the water-gas shift (WGS) reaction. The mechanistic insights provided into how surface hydroxyl groups contribute both to the electronic modulation of Pt and the dynamic restructuring of the oxide support are novel and significant. I find the study well-structured and impactful; however, I suggest the authors provide additional experimental or theoretical validation of the proposed Pt^0 -hydroxyl interaction mechanism and quantify the contribution of each step (CO adsorption vs. H_2O dissociation) to the overall WGS enhancement.*

Response: We would like to extend our sincere gratitude to the reviewer for the careful reading of our manuscript and the insightful comments. We have carefully addressed each point and made corresponding revisions to the manuscript. We hope that our responses and the revised manuscript will meet your expectations.

Comment 1: *Why is there such a large shift (~ 0.7 eV) in the O_{1s} binding energy of lattice oxygen ($O_{latt.}$) in TiO_2 -T-UC compared to TiO_2 -T-500 and TiO_2 -A? Please explain the origin of this significant difference.*

Response: We thank the reviewer for the valuable comment. We attribute the higher O_{1s} binding energy of the lattice oxygen in TiO_2 -T-UC, compared to TiO_2 -T-500 and TiO_2 -A, primarily to its fundamentally different crystal structure and local chemical environment of oxygen. Based on the combined analysis of XRD patterns (Figure R2-1) and relevant literature, the synthesized TiO_2 -T-UC support was identified as $H_2Ti_3O_7$, which exhibited the layered structure distinct from the anatase phase of TiO_2 -T-500 and

TiO₂-A (Tsai, C. C. et al. *Chem. Mater.* **2006**, 18, 367-373). The layered titanate structure inherently contained structural hydroxyl groups (-OH) and might incorporate interlayer water molecules. The O–H bonds within the structure exhibit the significant electronegativity difference. Hydrogen atoms withdrew electron density from oxygen atoms, thereby reducing the local electron density around the oxygen. This electronic environment results in the higher O 1s XPS binding energy compared to that of the Ti–O–Ti structure in anatase TiO₂. **The corresponding description has been added on Page 5, lines 8–11 in the revised manuscript (highlight in yellow). The corresponding reference has been cited as 38.**

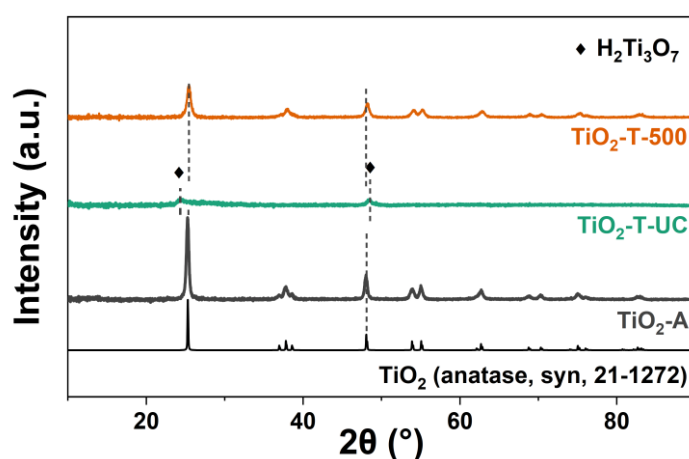


Figure R2-1. The XRD patterns of the TiO₂-A, TiO₂-T-UC and TiO₂-T-500 supports.

Comment 2: The statement “However, excessive surface hydroxyls caused the supported Pt species to be completely embedded in the TiO₂ support lattice during the WGS reaction, resulting in the loss of active metal sites (Supplementary Fig. S3 and S4)” requires quantitative data or literature support. Only two STEM images alone are not sufficient to support such a conclusion. A statistical comparison of the post-reaction STEM data between TiO₂-T-500 and the hydroxyl-rich sample would help clarify how the OH content influences the embedding or migration of Pt species.

Response: We thank the reviewer for the constructive comment. In order to provide the quantitative data, CO chemisorption experiments were performed on the spent catalysts

(Table R2-1). The results revealed that the 0.3Pt/(TiO₂-A) and 0.3Pt/(TiO₂-T-500) catalysts exhibited comparable CO uptake. In contrast, the 0.3Pt/(TiO₂-T-UC) catalyst exhibited the significant decrease in CO uptake, which indicated that the Pt species in 0.3Pt/(TiO₂-T-UC) were covered, thereby losing their CO adsorption capacity.

To further elucidate the origin of Pt species encapsulation by the TiO₂-T-UC support, the morphology of catalyst and the location of the Pt species were investigated using HAADF-STEM imaging. As shown in Figure R2-2a, the 0.3Pt/(TiO₂-T-UC) catalyst retained its nanotubular morphology even after the WGS reaction. Interestingly, the Pt species were distributed both on the inner and outer walls of the nanotubes, resulting in the non-uniform size distribution. In addition to the inadequate exposure of active sites for Pt particles located within the nanotube interior, those on the external surfaces were also found to be embedded within the tube walls. These encapsulated sites suppress the adsorption of CO molecules. When the support was calcined at 200 °C, the majority of the nanotubes were retained (Figure R2-2b). Similarly to the 0.3Pt/(TiO₂-T-UC) catalyst, Pt particles were incorporated into both the interior and the outer walls of these nanotubes, which hindered the exposure of active sites. In contrast to 0.3Pt/(TiO₂-T-UC), a portion of the nanotubes in the 0.3Pt/(TiO₂-T-200) catalyst collapsed, forming layered sheet-like structures. Although the Pt particles supported on these structures still exhibited a non-uniform size distribution, this morphological change increased the number of exposed Pt sites to absorb the CO molecules (Figure R2-3). These results demonstrated that the tubular morphology of TiO₂-T-UC led to the encapsulation of the supported Pt species. In contrast, the TiO₂-T-500 support completely collapsed into the sheet-like structure, which allowed the Pt species to become exposed after the reaction, thereby enhancing CO adsorption (Figure R2-4).

In summary, the tubular morphology of TiO₂-T-UC led to the encapsulation of the supported Pt species. Once the tubular structure of TiO₂ collapsed upon thermal treatment, the Pt species become exposed. Table R2-1 was shown as Table S1 in the supplementary information. Figure R2-2 and R2-3 has been shown as Figure S7 and S8 in the supplementary information. Figure R2-4 has been shown

as Figure 1h and 1i in the revised manuscript. The corresponding description has been added on Page 5, lines 16–22 in the revised manuscript (highlight in yellow).

Table R2-1. The CO uptake and Pt dispersion of 0.3Pt/(TiO₂-A), 0.3Pt/(TiO₂-T-UC) and 0.3Pt/(TiO₂-T-500) catalysts.

	CO uptake ($\mu\text{mol/g}$)	Dispersion
0.3Pt/(TiO ₂ -A)	7.4	48.1 %
0.3Pt/(TiO ₂ -T-UC)	2.9	18.8 %
0.3Pt/(TiO ₂ -T-500)	8.5	55.2 %

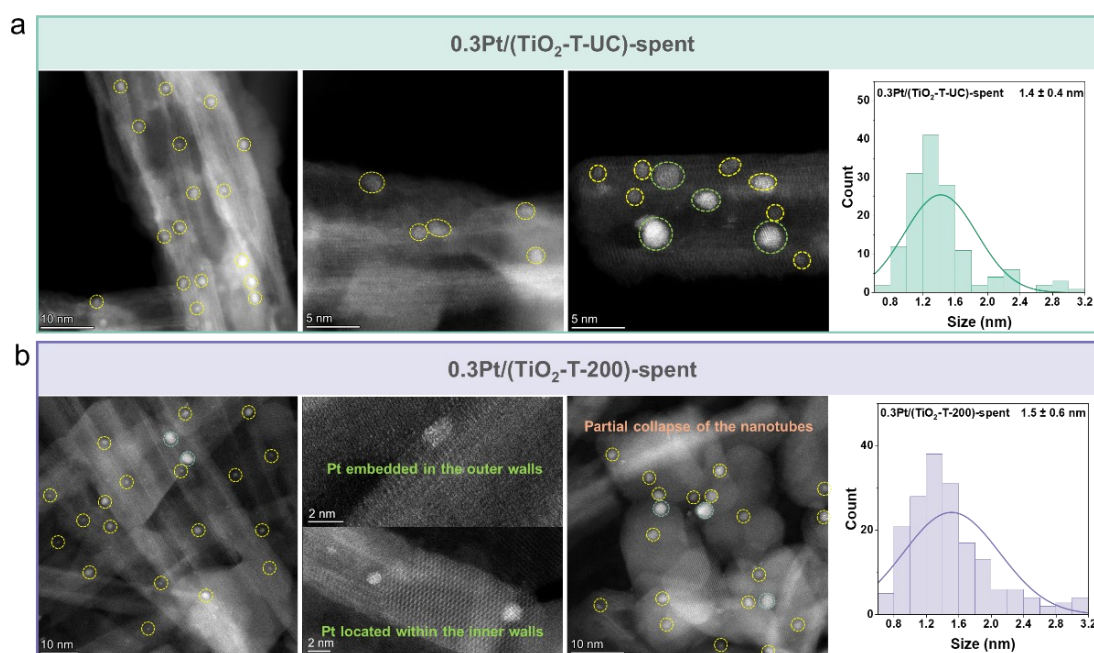


Figure R2-2. The aberration-corrected HAADF-STEM images and the size statistics of the spent (a) 0.3Pt/(TiO₂-T-UC) and (b) 0.3Pt/(TiO₂-T-200) catalyst.

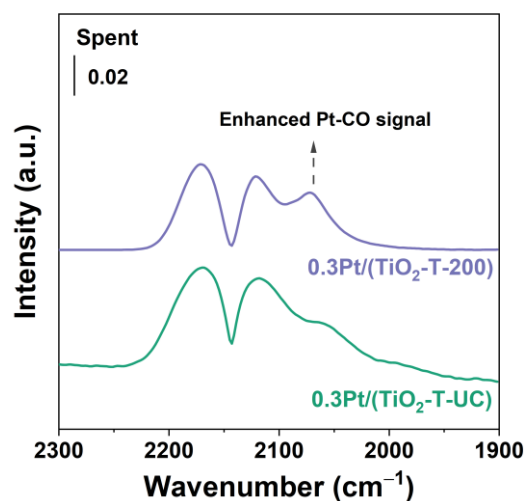


Figure R2-3. *In situ* DRIFTS spectra of the CO adsorption for the spent 0.3Pt/(TiO₂-T-UC) and 0.3Pt/(TiO₂-T-200) catalyst.

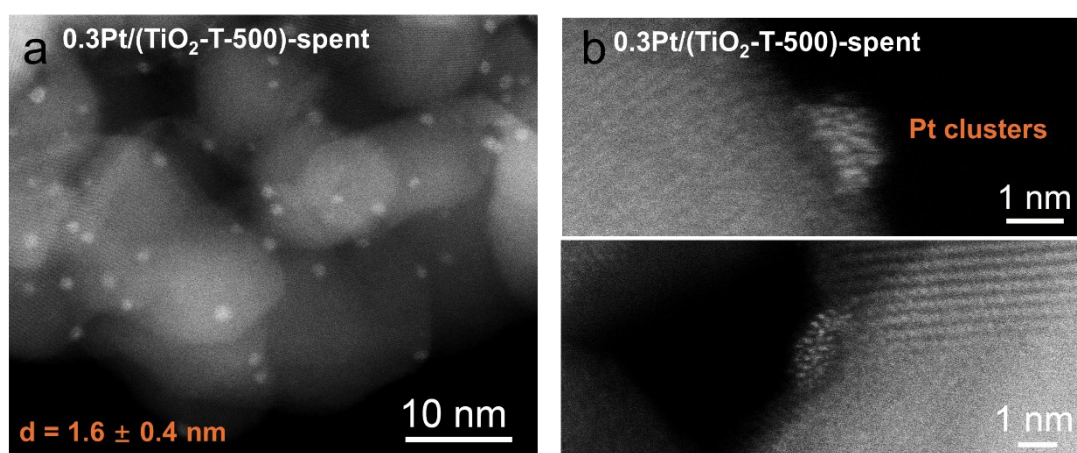


Figure R2-4. (a, b) Aberration-corrected HAADF-STEM images of the spent 0.3Pt/(TiO₂-T-500) catalyst.

***Comment 3:** The frequent mention of “OH content of TiO₂-T-500 (1.3 mmol g⁻¹)” gives the impression that the value may have been selectively emphasized. Please consider adjusting Figure 1 to avoid repetitive or potentially misleading data presentation.*

Response: We appreciate the reviewer for thoughtful suggestions. To avoid potential over-emphasis and misinterpretation by readers, we have removed the relevant data from Figure R2-5. Instead, the figure now simply annotated that the TiO₂-T-500 support

possessed the higher concentration of hydroxyl groups compared to TiO₂-A. **Figure R2-5 has been shown as Figure 1e in the revised manuscript and the corresponding description has been added on Page 7, lines 2–3 in the revised manuscript (highlight in yellow).**

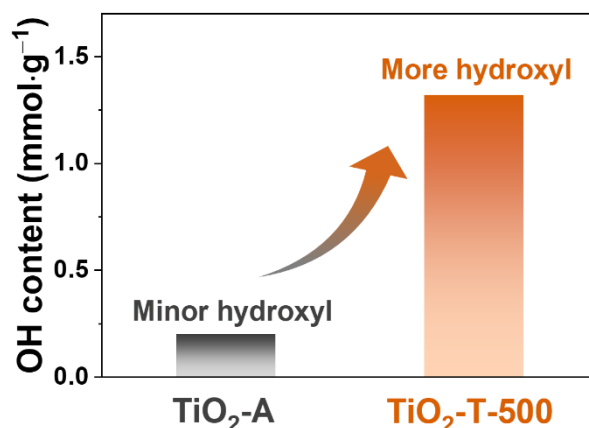


Figure R2-5. The hydroxyl content of TiO₂-A and TiO₂-T-500 supports determined from TGA measurements.

***Comment 4:** Please provide the WGS reaction conditions in the legend of Figure 1, especially those corresponding to Figure 1d. This information is necessary for clarity and reproducibility.*

Response: We sincerely thank the reviewer for the attentive comments. The reaction rates of the catalysts with supports calcined at different temperatures were presented in Figure R2-6. The tests were conducted at 250 °C under the reaction atmosphere composed of 2% CO/10% H₂O/N₂. **This information has now been added in the legend on Page 6, lines 4–6 (highlight in yellow).** We appreciate the reviewer for the valuable comments again.

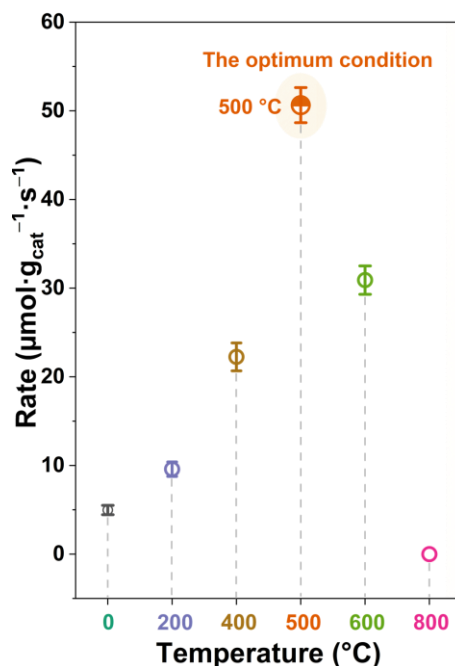


Figure R2-6. WGS reaction rate at 250 °C as the function of calcination temperature of TiO₂-T support, with the optimum at 500 °C. Reaction condition: 2% CO/10% H₂O/N₂.

***Comment 5:** The interpretation of Pt 4f spectra in Figure 2c requires further clarification. Specifically, 0.3Pt/(TiO₂-T-500) predominantly shows metallic Pt⁰ species, while 0.3Pt/(TiO₂-A) mainly exhibits Pt²⁺ or Pt⁴⁺ states. The orange peaks in both XPS spectra are currently labeled as Pt⁰, but the binding energy difference between them (~0.7 eV) is too large to be attributed to the same oxidation state. Please clarify the specific assignments of each deconvoluted peak.*

Response: We thank the reviewer for the valuable comments. According to the literature, in Pt 4f XPS spectra, peaks at binding energies above 74 eV were typically attributed to Pt⁴⁺, while those between 72-73 eV were generally assigned to Pt²⁺ (Lei, H. et al. *Angew. Chem. Int. Ed.* **2025**, e202509239; Wang, H. et al. *ACS Catal.* **2023**, 13, 10500–10510). This indicated that both catalysts contained Pt²⁺ species but no Pt⁴⁺. Furthermore, the peaks in the 70-71 eV range were commonly ascribed to Pt⁰, whereas those between 71-72 eV were usually attributed to Pt^{δ+} species (Tang, N. et al. *ACS*

Catal. **2024**, 14, 14297–14307; Liu, H. et al. *Nat. Common.* **2022**, 13, 5800; Li, K. et al. *ACS Appl. Mater. Interfaces* **2021**, 13, 52193–52201). Therefore, the Pt species in the 0.3Pt/(TiO₂-T-500) catalyst were predominantly in the metallic Pt⁰ state, while in the 0.3Pt/(TiO₂-A) catalyst, the Pt species existed mainly as Pt^{δ+} species. **The reassigned XPS spectra was now presented in Figure R2-7 and have been showed in the revised manuscript as Figure 2b. The corresponding description has been added on Page 8, lines 7–10 in the revised manuscript (highlight in yellow) and the relevant references have been cited as 41–43.**

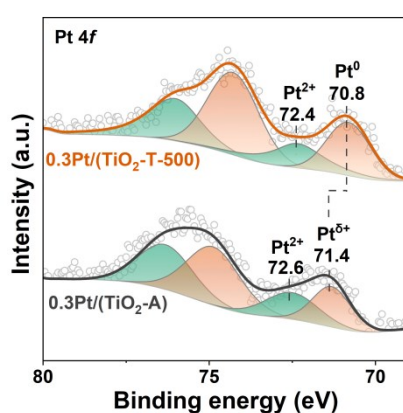


Figure R2-7. *Ex situ* Pt 4f XPS for 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) samples after H₂ pretreatment.

Comment 6: *The phrase “the formation of electron-rich Pt⁰ species” is somewhat inappropriate. It may be unnecessary to emphasize “electron-rich,” as the main point is the formation of metallic Pt⁰ species. There is no clear evidence that these Pt⁰ species are more electron-enriched than typical Pt⁰, as indicated by the XPS results in Figure 2c.*

Response: We thank the reviewer for this professional comment. **In the revised manuscript, we have amended the phrasing to “Pt⁰ clusters”, including both the title and the main text.**

Comment 7: *In the in situ DRIFTS spectra of the 0.3Pt/(TiO₂-A) catalyst, why does the CO₂ signal appear at the very beginning (Figure S14), while for 0.3Pt/(TiO₂-T-500) the signal is much weaker or absent? This suggests that the Pt^{δ+} – OH – Ti⁴⁺ interfacial sites in the former are highly reactive initially, rapidly oxidizing CO to CO₂. Please discuss this. Also, how should we interpret the volcano-shaped curve of CO₂ formation in Figure 2i? A clear explanation is needed in the text.*

Response: We thank the reviewer for the professional comments. Since the generation of H₂ could not be detected by *in situ* DRIFTS, we employed mass spectrometry to monitor the catalytic behavior during the heating process of the water gas shift (WGS) reaction. As shown in Figure R2-8, both catalysts exhibited rapid consumption of reactant CO molecules before the occurrence of the WGS reaction. However, a notable difference was observed. The 0.3Pt/(TiO₂-A) catalyst produced only CO₂ without H₂, indicating that abundant surface oxygen on the catalyst facilitated the reaction with CO at relatively low temperatures to form CO₂, rather than involving the Pt^{δ+}–OH–Ti⁴⁺ structure. In contrast, the 0.3Pt/(TiO₂-T-500) catalyst generated both CO₂ and H₂, demonstrating that the reaction between Pt⁰-OH and CO occurred. Therefore, the gaseous CO₂ signal detected by *in situ* DRIFTS during the CO adsorption for 0.3Pt/(TiO₂-A) originated from CO₂ formed by the reaction between CO and surface oxygen.

To eliminate the influence of surface oxygen on the 0.3Pt/(TiO₂-A) catalyst, multiple CO adsorption and purging experiments were conducted during *in situ* DRIFTS testing. As shown in Figure R2-9, the gaseous CO₂ signal progressively weakened with successive CO adsorption cycles and nearly disappeared after the fourth cycle, indicating that surface oxygen was effectively removed. Therefore, after four CO adsorption cycles, switching experiments involving CO and Ar or Ar+H₂O were performed again. As shown in Figure R2-10, at 250 °C, the CO adsorption peak on Pt^{δ+} species of 0.3Pt/(TiO₂-A) gradually diminished under Ar purging, and upon re-adsorption of CO, the CO signal at 2080 cm⁻¹ re-emerged. Subsequently, as shown in Figure R2-11a, upon switching to the Ar+H₂O atmosphere, the CO adsorption peak

similarly diminished gradually, with the decay rate consistent with that observed under pure Ar purging. Meanwhile, the concurrent decrease in the gaseous CO₂ signal occurred. In contrast, for the 0.3Pt/(TiO₂-T-500) catalyst, the CO signal corresponding to Pt⁰ species decayed significantly faster under the Ar + H₂O atmosphere than under pure Ar, accompanied by the gaseous CO₂ signal that initially increased and subsequently decreased (Figure R2-11b). This indicated that CO adsorbed on Pt⁰ sites reacted with H₂O, generating additional CO₂. As the adsorbed CO on Pt⁰ was progressively consumed, the production of CO₂ gradually diminished and eventually vanished upon complete purging. These results collectively demonstrated the superior ability of metallic Pt⁰ in activating CO molecules.

Figure R2-9 has been shown as Figure S21 in the supplementary information. Figure R2-10 has been shown as Figure S22 in the supplementary information. Figure R2-11 has been shown as Figure 2h and 2i in the revised manuscript. The corresponding description has been added on Page 10, lines 22–29 and Page 11, lines 1–5.

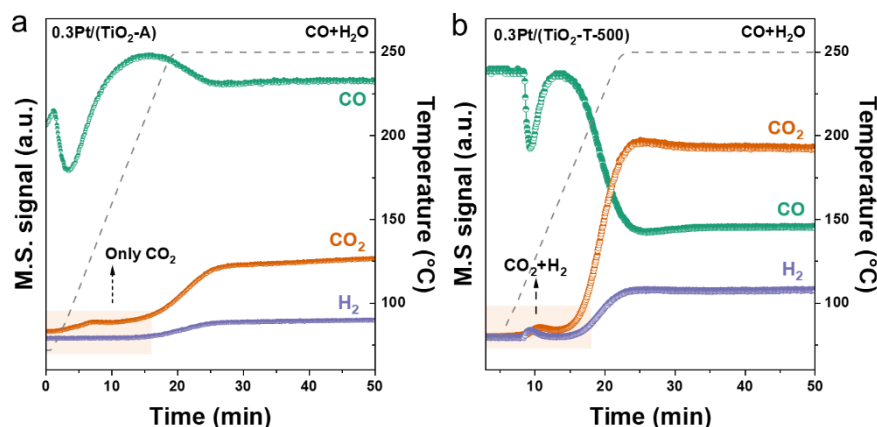


Figure R2-8. The catalytic behavior of (a) 0.3Pt/(TiO₂-A) and (b) 0.3Pt/(TiO₂-T-500) during the heating process of the water gas shift (WGS) reaction.

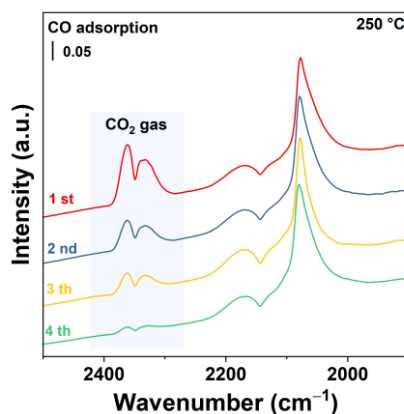


Figure R2-9. Four cycles of CO adsorption experiments conducted under *in situ* DRIFTS spectroscopy of 0.3Pt/(TiO₂-A) catalyst at 250 °C.

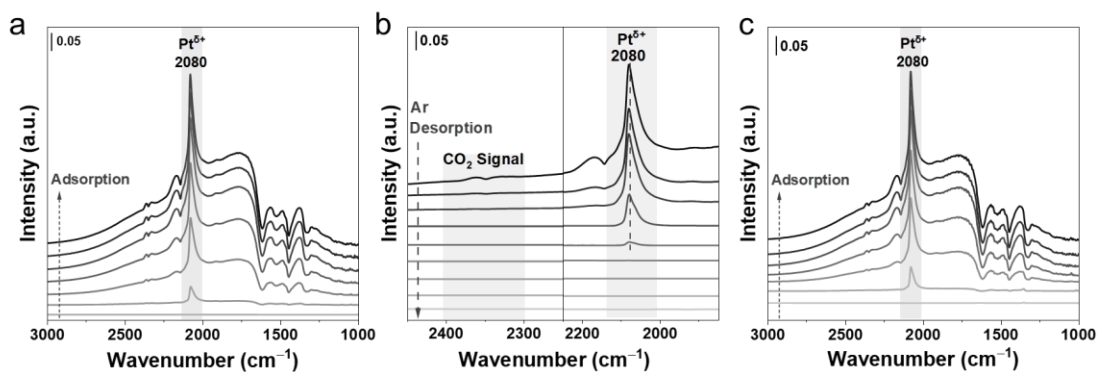


Figure R2-10. *In situ* DRIFTS spectra of (a) CO adsorption, (b) Ar desorption and (c) CO re-adsorption over 0.3Pt/(TiO₂-A) catalyst at 250 °C.

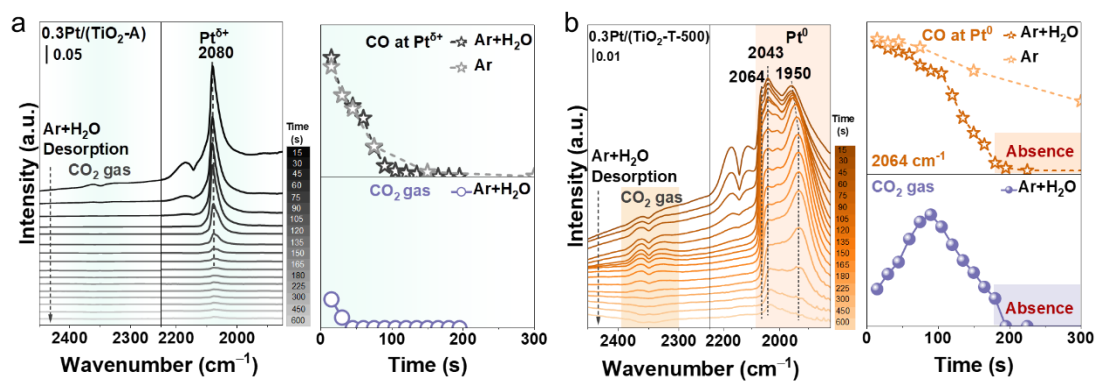


Figure R2-11. (a, b) *In situ* DRIFTS spectra of Ar+H₂O purging over (a) 0.3Pt/(TiO₂-A) and (b) 0.3Pt/(TiO₂-T-500) at 250 °C. Right: the intensity of CO adsorption on Pt sites upon Ar+H₂O and Ar purging as function of time.

Comment 8: *As the authors state, “These results revealed that the electron-rich Pt^0 closer to the interface was more active in activating CO molecules.” However, is it possible that the “ Pt^0 near the interface” is in fact the $Pt\delta^+$ species located at the $Pt\delta^+-OH-Ti^{4+}$ interface, rather than fully metallic Pt^0 ? This may involve dynamic transformation between $Pt\delta^+$ and Pt^0 driven by OH consumption and water dissociation during the reaction. Consider using DFT simulations and valence state calculations to further support this hypothesis.*

Comment 9: In Figure 3b, the CO₂ and H₂ signals during the second CO injection are significantly weaker than those during the first injection. This indicates that the interfacial OH species regenerated via water dissociation are less abundant than the native surface OH species originally present on the catalyst. What could be the reason? Is it due to structural changes in the catalyst or sintering of the Pt nanoparticles?

Response: We thank the reviewer for the attentive comments. The substantial and sustained production of CO₂ and H₂ observed in the first stage (0–65 min) was attributed to the simultaneous introduction of CO and H₂O. During this stage, H₂O underwent continuous dissociation, while CO reacted to form CO₂ and H₂. In the fifth stage (the second CO injection), however, only CO was introduced, which could only react with the hydroxyl groups generated during the pre-hydroxylation process of the previous stage. Without additional replenishment of H₂O, the production of CO₂ and H₂ in this stage was consequently relatively limited. **We have annotated the Stage I reaction atmosphere (CO + H₂O) in Figure R2-13 to provide readers with a clearer understanding of the gas conditions during each stage. The Figure R2-13 has been updated as Figure 3b in the revised manuscript.**

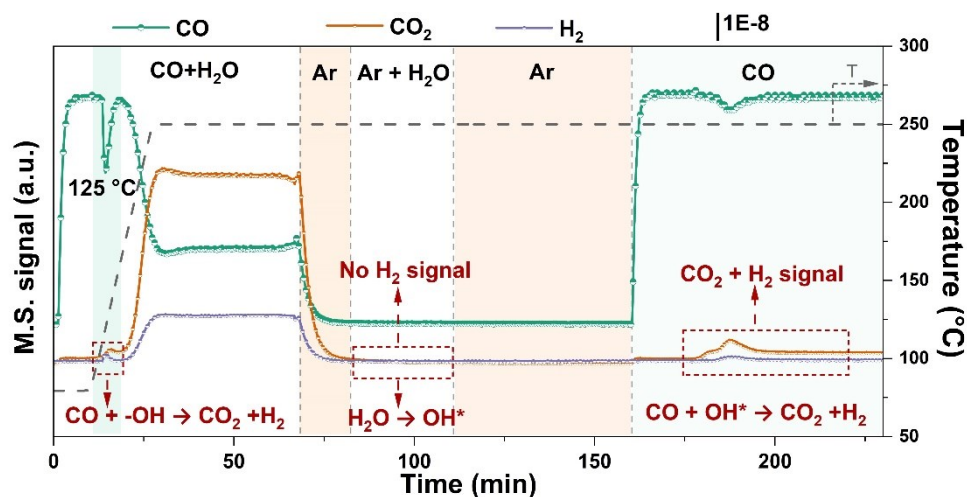


Figure R2-13. (a, b) *In situ* DRIFTS spectra of Ar+H₂O purging over (a) 0.3Pt/(TiO₂-A) and (b) 0.3Pt/(TiO₂-T-500) at 250 °C. Right: the intensity of CO adsorption on Pt sites upon Ar+H₂O and Ar purging as function of time.

Comment 10: After the long-term stability test, how does the Pt 4f XPS spectrum of the 0.3Pt/(TiO₂-T-500) catalyst compare with the fresh sample? Are there any notable changes in the oxidation state of Pt species?

Response: We thank the reviewer for the constructive comments. As shown in Figure R2-14a, the Pt species in the fresh 0.3Pt/(TiO₂-T-500) catalyst existed as Pt²⁺. However, after the long-time stability test, the catalyst was predominantly composed of Pt⁰ species, indicating the reduction of Pt species during the reaction (Figure R2-14b). Notably, the H₂-treated 0.3Pt/(TiO₂-T-500) catalyst exhibited peaks at 70.8 eV and 72.2 eV, attributed to Pt⁰ and Pt²⁺ species respectively (Figure R2-15). After the long-term stability test, the Pt⁰ peak further shifted to the lower binding energy of 70.4 eV and the Pt²⁺ species were further reduced to Pt^{δ+}, appearing at 71.2 eV. These results demonstrated that the Pt species in 0.3Pt/(TiO₂-T-500) also underwent further reduction during the WGS reaction. **Figure R2-14a has been shown as Figure S16a in the supplementary information. Figure R2-14b has been shown as Figure S35 in the supplementary information. Figure R2-15 has been shown as Figure 2b in the revised manuscript. The corresponding description has been added on Page 16, lines 22–24 (highlight in yellow).**

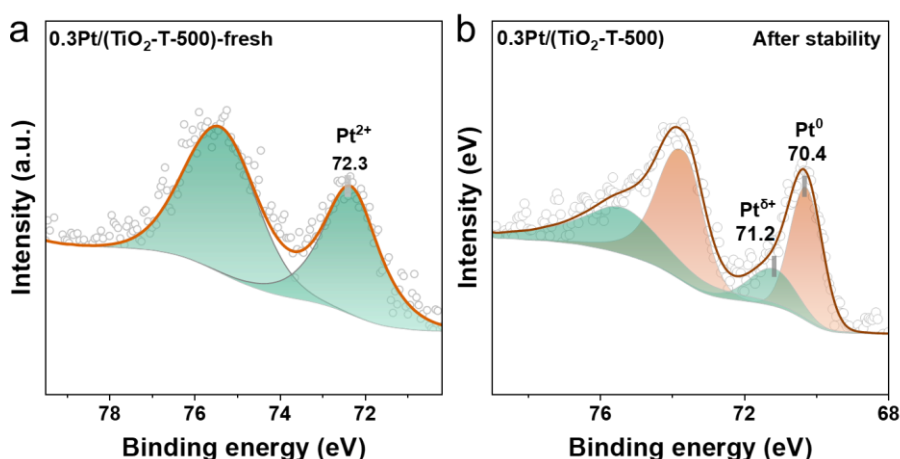


Figure R2-14. The Pt 4f XPS spectra of the 0.3Pt/(TiO₂-T-500) catalyst (a) before reaction and (b) after the 60 h stability test.

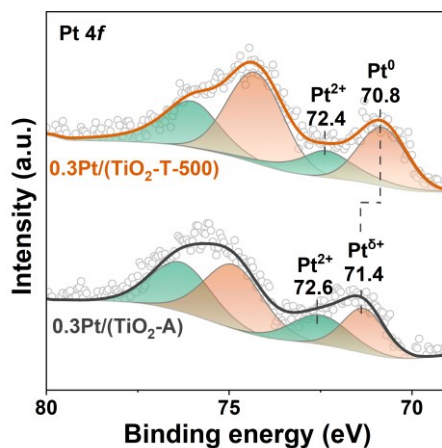


Figure R2-15. The Pt 4f XPS for 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) samples after H₂ pretreatment.

***Comment 11:** Based on the above concerns, I would like to recommend the publication of this work in Nature Communications.*

Response: We thank the reviewer for the constructive and positive comments on our work, which were very valuable for improving the quality of our manuscript. We have thoroughly revised the manuscript accordingly. We hope that the revised manuscript is now suitable for publication in *Nature Communications*.

Responses to the Reviewers' Comments and the Corresponding Revisions

Reviewer #1:

Comment 1: *The authors have revised their manuscript and the central idea remains that tuning the hydroxyl population on TiO₂ has two coupled effects: it biases reduced Pt toward a more metallic electronic character that strengthens CO binding and activation, and it promotes oxygen-vacancy generation during turnover in a way that enhances water activation. The overall narrative is coherent and, at a high level, plausible for WGS on reducible oxides. I also find the materials handle conceptually appealing and potentially useful: deliberate hydroxyl regulation via a titanate-derived route and controlled calcination to reach an apparent optimum hydroxyl content at 500 °C. The optimization itself is one of the more convincing aspects of the work, particularly because the authors acknowledge that excessive hydroxylation can lead to Pt encapsulation in the uncalcined nanotubular precursor and they relate that state to suppressed CO uptake and suppressed activity for the hydroxyl-rich TiO₂-T-UC sample, rather than relying on microscopy alone.*

The revision is strengthened by the addition of TOFs, chemisorption-derived dispersions, reaction orders, and additional spectroscopy. The reported performance differences are nontrivial and persist after normalization to accessible Pt sites as the authors define them. In particular, the revised manuscript reports a rate of 62.7 mol_{CO} g_{Pt}⁻¹ h⁻¹ and a TOF of 6.15 s⁻¹ for 0.3Pt/(TiO₂-T-500) at 250 °C, compared with 31.4 mol_{CO} g_{Pt}⁻¹ h⁻¹ and 3.54 s⁻¹ for 0.3Pt/(TiO₂-A). The authors also state that the two catalysts show similar apparent activation energies and therefore likely follow the same overall mechanism, which is an important and appropriate framing that supports a rate-enhancement claim rather than a claim of a fundamentally new pathway.

Response: We thank the reviewer for the valuable comments, which have undoubtedly helped improve the logical flow and evidential support of the manuscript.

Comment 2: *My main concern is that key mechanistic statements still read more certain than the current evidence warrants. In several places the manuscript relies on*

chains of observations that are mutually consistent but not uniquely discriminating, and the conclusions drift from “consistent with” toward “demonstrated”. This is most evident in the assignment of Pt^0 versus $Pt^{\delta+}$. The assignment is still based primarily on CO-DRIFTS band positions and small XPS shifts, supported by an in situ Pt L₃-edge XANES comparison acquired under H₂ pretreatment. The authors themselves note limitations here, including signal-to-noise constraints at 0.3 wt% and the fact that the measurement conditions are not operando WGS. The chemical rationale offered for the DRIFTS shift (reduced back-donation on electron-deficient $Pt^{\delta+}$ leading to higher CO stretching frequency) is reasonable, but it does not resolve the core ambiguity: on Pt/TiO₂, CO band positions can be influenced by oxidation state as well as particle size, coverage, ensemble effects, and interfacial geometry.

Response: We thank the reviewer for this important comment, which helps us sharpen the scientific narrative. We have revised the manuscript throughout to more carefully reflect the supportive, rather than demonstrative, role of the spectroscopic data. Key changes include:

1. Replacing verbs like “demonstrated” or “proved” with “**suggested,**” “**indicated,**” “**pointed to**”, “**was associated with**”, or “**was consistent with**” in sections discussing Pt electronic state. The corresponding changes have been made on page 9, line 14 and 28; page 12, line 15 and 25 of the revised manuscript (highlight in yellow).
2. We added explicit discussions on the inherent limitations of the characterization techniques employed. Specifically, for the *in situ* XANES data acquired under H₂, we now clearly state on Page 9, lines 16–23 (highlight in yellow): “**Considering the poor signal-to-noise ratio and interference from H* adsorption under H₂ atmosphere, this result could not fully represent the exact state of Pt species under the WGS reaction, which could only imply a stronger tendency for Pt species in 0.3Pt/(TiO₂-T-500) to be in the metallic state compared to that in 0.3Pt/(TiO₂-A).**” Regarding the *in situ* DRIFTS results, we have incorporated the necessary caution on Page 9, line 29 and Page 10, lines 1–3: “**While we acknowledge that band positions can also be influenced by factors such as particle size and CO coverage, the observed trend**

across the sample series, combined with the supporting HAADF-STEM images, XPS and XANES data, leads us to attribute the dominant effect to the modification of the Pt electronic state."

3. Finally, our overall conclusion has been reframed with appropriate caution on Page 10, lines 3–5 (highlight in yellow): "Overall, the observed trends were consistent with a mechanistic interpretation wherein support hydroxylation promoted the formation of Pt species with the more metallic electronic character."

Overall, incorporating the reviewer's valuable comment has allowed us to present a more coherent and cautious mechanistic discussion that connects the experimental trends while acknowledging the boundaries of the current evidence, which has greatly strengthened the manuscript.

Comment 3: The authors have added post-reaction HAADF-STEM statistics and CO chemisorption on spent catalysts, and the mean sizes appear similar after reaction (about 1.6–1.7 nm). However, the chemisorption table shows dispersions that are not identical (about 55.2% for TiO₂-T-500 versus 48.1% for TiO₂-A in the provided dataset). With such low metal loading, this difference is not obviously negligible, and it matters because the chemisorption-derived site count is being used both to normalize activity (TOF) and to support mechanistic interpretation. If the claim is that intrinsic site quality changes, the manuscript would benefit from a clearer demonstration that CO uptake and dispersion estimates are robust and not systematically biased by support-dependent spillover, readsorption, or differences in CO binding strength that could affect the inferred site count. At minimum, the authors should temper conclusions that hinge critically on dispersion-derived normalization and explicitly discuss the uncertainty and assumptions.

Response: We sincerely thank the reviewer for the valuable comment regarding the robustness of CO chemisorption for site counting. We fully acknowledge the reviewer's concern that factors such as support-dependent spillover, readsorption dynamics, or differences in CO binding strength could, in principle, introduce systematic bias into

the measured CO uptake and the derived dispersion, especially at such low metal loadings. However, we wish to emphasize that even if the absolute site counts are subject to such a bias, the relative comparison between the two catalysts remains valid and highly informative. The key observation is that while the dispersion of 0.3Pt/TiO₂-T-500 is only ~10% higher than that of 0.3Pt/TiO₂-A, its TOF is nearly doubled. This pronounced discrepancy strongly suggests an intrinsic difference in the catalytic quality of the Pt sites induced by the TiO₂ supports.

We fully concur that when discussing TOF values, the potential interference of other factors on dispersion estimates must be carefully considered. However, it is extremely difficult to precisely eliminate the contribution of these intertwined factors to the total CO signal with currently accessible techniques. Therefore, in line with the reviewer's suggestion for a minimum revision, we have chosen to **fundamentally reframe our discussion and conclusions** to explicitly discuss the uncertainty and assumptions, rather than relying on potentially ambiguous additional measurements. Our revisions focus on the following key points, which we believe strengthen the manuscript significantly:

1. **Explicit discussion of limitations:** We have added a dedicated discussion explicitly stating the assumptions and potential limitations of CO chemisorption for site quantification in the section discussing dispersion and TOF.
2. **Tempered language and focus on trends:** We have revised the text to temper the conclusions regarding TOF: “**Assuming that the CO chemisorption measurements were comparable for both catalysts, the observation of a markedly higher TOF indicated a potentially greater intrinsic activity of the active sites on the TiO₂-T-500 support**” The mechanistic claim was now pivoted to emphasize the relative trend of activity enhancement.

The corresponding description has been revised on Page 8, lines 18–23 and Page 18, lines 10–17 of the revised manuscript (highlight in yellow).

Comment 4: *The second major mechanistic pillar, that hydroxyl consumption by CO generates oxygen vacancies that become the dominant loci for water dissociation, is*

intriguing and not unreasonable, but it is still somewhat over-interpreted. The revision links vacancy formation to EPR changes after WGS and to Ti-related spectral features interpreted as vacancy signatures, and then links “more vacancies” to “better water activation” through reaction orders and a kinetic isotope effect. The reported KIE of roughly 1.65 upon substituting D₂O for H₂O indicates that O–H activation is kinetically relevant, but it does not uniquely identify the elementary step, nor does it establish that vacancies created specifically by hydroxyl consumption are the operative active sites for that step. Moreover, the manuscript also states that hydroxyl signals increase after reaction due to water dissociation and adsorption, implying that hydroxyls are continuously regenerated during WGS. In such a dynamic system it is important to distinguish hydroxyls as spectators, hydroxyls as true intermediates on the main catalytic cycle, and hydroxyls as part of a parallel redox process that modulates defect populations without being kinetically dominant. At present the evidence is consistent with the proposed picture, but causality remains incompletely established. I would therefore ask for at least one genuinely discriminating experiment or, alternatively, a more conservative mechanistic framing. Suitable discriminating experiments could include isotope tracing (for example H₂¹⁸O, C¹⁸O, or ¹⁸O-enriched supports) to determine whether oxygen in CO₂ derives predominantly from water, lattice oxygen, or hydroxyl oxygen under the relevant catalyst states, or operando measurements that correlate a quantitative defect metric with instantaneous rate under controlled perturbations.

Response: We sincerely thank the reviewer for the insightful and constructive critique of the mechanistic section of our manuscript. In response to this concern, we have conducted H₂¹⁸O isotopic tracing experiments by *in situ* DRIFTS and mass spectrometry. Firstly, the catalyst was first subjected to CO-TPR aimed at removing hydroxyl groups, as evidenced by the simultaneous generation of H₂ and CO₂ around 150 °C (Figure R1a). Subsequently, H₂¹⁸O was introduced for 20 minutes. Upon switching back to the CO flow, signals for C¹⁶O¹⁸O and C¹⁸O¹⁸O were detected (Figure R1b). This observation suggested the possibility that H₂O dissociated on the catalyst

surface and the incorporation of its oxygen into the product. Meanwhile, the presence of $C^{16}O^{16}O$ was indicative of reactions between CO and surface oxygen atoms or the initially engineered hydroxyl groups. Moreover, during the same experiment by *in situ* DRIFTS, $C^{16}O^{18}O$ was also detected, providing further evidence for the dissociation of H_2O on the catalyst surface (Figure R2). **The corresponding description has been added on page 15, line 30 and page 16, line 1–5 in the revised manuscript (highlight in yellow).** Figure R1 and R2 has been added as Figure S31 and S32 in the revised supplementary information.

However, while this result supported the occurrence of the water activation step, it did not uniquely identify the atomic-scale origin of the active site whether it belongs exclusively to vacancies generated via hydroxyl consumption. Therefore, we have chosen to adopt the reviewer's alternative suggestion and have implemented a thorough revision towards a more conservative mechanistic framing. Key changes in the revised manuscript include:

- 1. Reframing the narrative:** The entire mechanistic discussion has been shifted from a definitive "demonstration" to an inferential "proposal of a self-consistent model." The core narrative now uses phrasing such as **"A plausible explanation for this difference was that..."**, **"The above results provided insight into the correlation between..."**, **"A reasonable model to explain these observations was that..."** and so on.
- 2. Stating observed correlations, weakening causal interpretation:** We have replaced statements that implied a mechanistic direction with direct statements of the observed correlations. For instance, we now state: **"The above results provided insight into the correlation between the consumption of surface hydroxyl groups by CO and the facilitated formation of oxygen vacancies."**, **"The consumption of hydroxyl groups by CO, whether initially present or formed during the WGS reaction, suggested a link to the formation of oxygen vacancies."** and **"0.3Pt/(TiO₂-T-500) with more defects exhibited the greater capacity for H₂O dissociation"**. The corresponding description has been revised on page 14, line 4; page 15, lines 9–10 and 20–23 and page 16, lines 5–8 in the revised manuscript (highlight in yellow).

3. **Explicit acknowledgement of limitations and system dynamics:** A dedicated paragraph has been added outlining the limitations of the current evidence, including the inability of the isotopic experiment to pinpoint the origin of the active site. Furthermore, given that hydroxyl groups are dynamic under reaction conditions, their precise roles cannot be unambiguously assigned to individual steps, underscoring the complexity of the working catalyst surface. **The corresponding description has been added on page 16, lines 8–11 in the revised manuscript (highlight in yellow).**

4. **Revised conclusion:** The conclusion has been refocused from "elucidating a mechanism" to "establishing a strong correlation and proposing a rational framework that links hydroxyl chemistry to enhanced water activation capability." We now conclude that **the hydroxyl-tuning strategy results in an optimal catalyst that concurrently exhibits higher oxygen vacancy signatures and a greater capacity for water activation. The corresponding description has been added on page 16, lines 11–14 in the revised manuscript (highlight in yellow).**

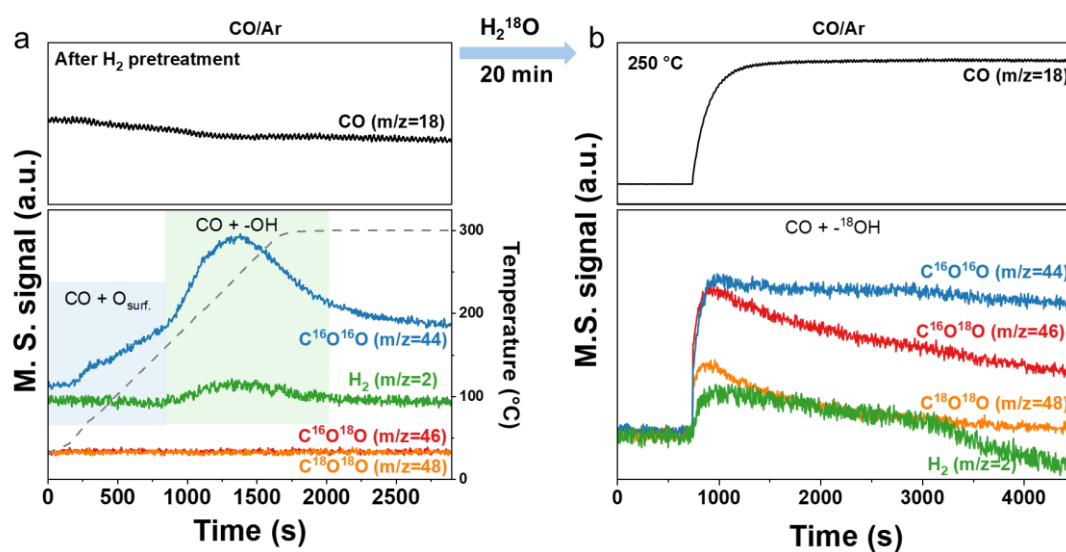


Figure R1 (a) CO-TPR profiles of 0.3Pt/(TiO₂-T-500) after H₂ pretreatment. The grey dotted line represents the working temperature. (b) The generation of CO₂ when injecting CO/Ar after H₂¹⁸O dissociation at 250 °C.

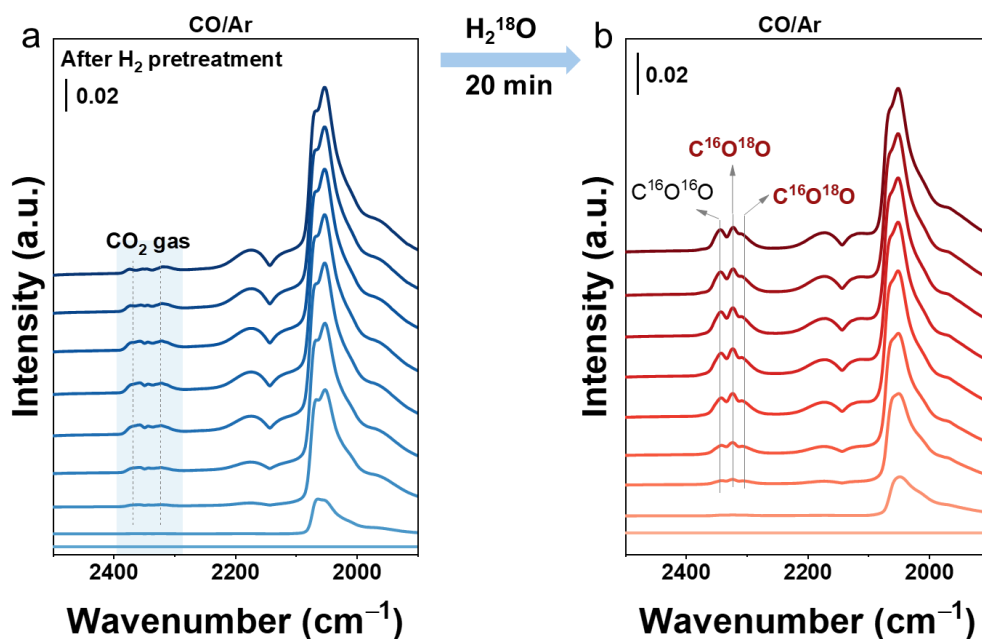


Figure R2 *In situ* DRIFTS spectra of the CO adsorption for 0.3Pt/(TiO₂-T-500) after (a) H₂ pretreatment and (b) subsequent H₂¹⁸O treatment.

Comment 5: *A further concern is a plausible confound that is not convincingly excluded. The hydroxylated support is produced using concentrated NaOH and acid washing to form titanate intermediates before calcination. Trace residual sodium or chloride can measurably alter TiO₂ surface chemistry, hydroxyl stability, vacancy formation energetics, and WGS kinetics, and these effects could mimic a “hydroxyl regulation” trend. Given that the manuscript compares performance against literature benchmarks that include alkali-modified Pt/TiO₂ systems, it becomes especially important to demonstrate that residual promoters are not contributing to the observed behavior. If not already provided, elemental analysis such as ICP/OES (or similarly convincing quantification) for Na and Cl on the final catalysts should be included, along with a brief discussion of detection limits.*

Response: We thank the reviewer for raising this critical point. This following study examined the residual content of Na and Cl on the catalyst surface and ruled out their contribution to the activity enhancement.

1. Regarding the residual Na:

(1) For the analysis of Na residues, ICP-MS was employed (Table R1), which showed that 0.3Pt/(TiO₂-T-500) catalyst contained 0.49 wt.% Na. In contrast, the Na content in the 0.3Pt/(TiO₂-A) catalyst was below the detection limit (<50 ppt). Relative to the mass of the catalyst, this corresponded to a residual Na level of less than 0.00125 wt.%. The results indicated that Na indeed remained on the catalyst surface during the synthesis process.

(2) To minimize residual ions, we implemented an intensive washing protocol using 5 L of high-purity water for the titanate precursor. In the resulting catalyst, 0.3Pt/(TiO₂-T-wash-500), Na content was below the detection limit of ICP-MS, thus suggesting the efficacy of extensive washing for Na removal (Table R1). Notably, the washed catalyst, 0.3Pt/(TiO₂-T-wash-500), matched the original 0.3Pt/(TiO₂-T-500) both in its high WGS activity and in its ATR-IR hydroxyl signal profile, with both being superior to the 0.3Pt/(TiO₂-A) catalyst (Figure R3a and b). This provided the understanding that the activity enhancement was not attributable to Na residues.

(3) Furthermore, we also found that the residual Na content on the surface of the 0.3Pt/(TiO₂-T-400) catalyst was at the similar level as that on the 0.3Pt/(TiO₂-T-500) catalyst (Table R1). However, the catalytic reaction rate of 0.3Pt/(TiO₂-T-500) was twice that of 0.3Pt/(TiO₂-T-400) (Figure R4). This further indicated that residual Na was not the source of the enhancement in catalytic performance.

2. Regarding the residual Cl:

(1) We clarify that standard ICP-OES/MS is not the optimal technique for accurate quantification of chloride anions due to poor sensitivity in the vacuum UV region and significant spectral interferences, which preclude reliable detection at the low ppm/ppb levels relevant here. Thus, we could only perform a semi-quantitative assessment and trend analysis of the Cl residues on the surfaces of different catalysts using Cl 2p XPS. As shown in Table R1, Figure R3c and d, the signal of residual Cl on the catalyst surfaces was weak. As indicated by semi-quantitative XPS analysis, the difference in Cl content between 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) was within 0.2 wt.%.

Therefore, the observed activity difference between the two catalysts could not be attributed to residual Cl.

(2) In addition, while chlorine residues can aid in catalyzing some organic reactions for oxide-supported catalysts, for gas-phase reactions, chlorine species on the catalyst surface typically exhibit strong binding to metal sites, thereby hindering the adsorption of reactant molecules and suppressing catalytic activity (Gracia, F. J. et al. *J. Catal.* **2002**, 209, 341–354; Bartholomew, C. H. et al. *Appl. Catal. A: Gen.* **2001**, 212, 17–60; Jardim, E. O. et al. *J. Colloid Interface Sci.* **2015**, 443, 45–55). Therefore, the presence of Cl residues on the catalyst surface was not considered to be the source of enhanced catalytic activity.

In summary, the above exclusion of the influence of Na and Cl on catalytic performance has been added on page 17, lines 10–11 and page 18, lines 1 in the manuscript and page 47, lines 5–24 in the Supporting Information (highlight in yellow). The Table R1 has been added as Supplementary Table 3 in the Supporting Information. The Figure R3b, c, d has been added as Supplementary Fig. S34 in the Supporting Information. In addition, we have added the ICP-MS test method and a brief analysis of the detection limits on page 20, lines 19–27 in the manuscript (highlight in yellow).

Table R1. Residual Na and Cl content on the catalyst surface

Samples	Sodium residue (wt.%) ^a	Chloride residue (wt.%) ^b
0.3Pt/(TiO ₂ -T-500)	0.49	0.95
0.3Pt/(TiO ₂ -A)	<LOD	0.81
0.3Pt/(TiO ₂ -T-500)-wash	<LOD	1.07
0.3Pt/(TiO ₂ -T-400)	0.59	—

^a Determined by ICP-MS; LOD: limit of detection. The method LOD was 0.00125 wt.% for Na.

^b Determined by XPS;

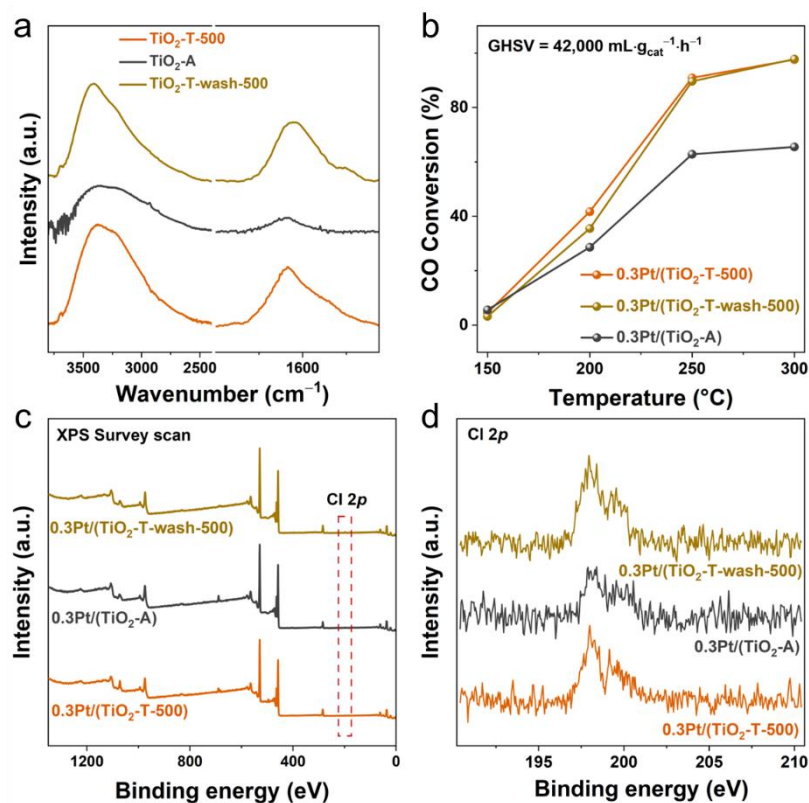


Figure R3. (a) ATR-IR of TiO₂-T-500, TiO₂-A and TiO₂-T-wash-500. (b) Temperature-dependent CO conversion of 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) and 0.3Pt/(TiO₂-T-wash-500) catalysts for the WGS reaction. (c) The XPS survey and (d) Cl 2p spectra of the fresh 0.3Pt/(TiO₂-T-500), 0.3Pt/(TiO₂-A) and 0.3Pt/(TiO₂-T-wash-500) catalysts.

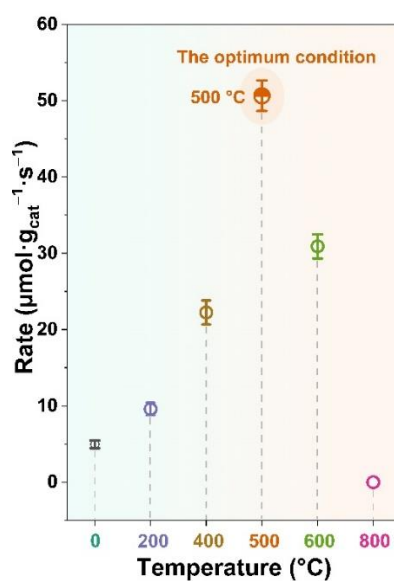


Figure R4 WGS reaction rate at 250 °C as the function of calcination temperature of TiO₂-T support, with the optimum at 500 °C. Reaction condition: 2% CO/10% H₂O/N₂.

Comment 6: *I would also like to comment on the balance of the cited literature. Approximately half of the references in the manuscript are authored exclusively by Chinese research groups, which is unusually high for a general catalysis paper in Nature Communications. Many of these citations are used to support broad and well-established concepts such as metal–support interactions, hydroxyl effects, and defect-mediated water activation. At the same time, several foundational and widely recognized contributions from the broader international literature are under-represented or absent. This includes classical frameworks for metal–support interaction and reducible oxide chemistry, as well as established kinetic and mechanistic studies of WGS on Pt/TiO₂. The current citation pattern risks giving the impression that recent regional literature defines these concepts, rather than placing the present work within the full historical and international context of the field. Such a balance is problematic from a scholarly perspective, and I would strongly recommend that the author balance the references to better reflect the breadth of prior work and to clearly acknowledge the foundational studies on which this manuscript builds.*

Response: We thank the reviewer for this professional comment. Accordingly, we have significantly revised the reference list. Key foundational and international papers and reviews on **water gas shift reaction** (Ratnasamy, C et al. *Catal. Rev.* **2009**, 51, 325–440); **the various reactions employed in the Pt/TiO₂ system** (An, K. et al. *Nano Lett.* **2014**, 14, 4907–4912; Kattel, S. et al. *J. Catal.* **2016**, 343, 115–126; Bonanni, S. et al. *J. Am. Chem. Soc.* **2014**, 136, 8702–8707; Kalamaras, C. M. et al. *J. Catal.* **2009**, 64, 117–129) **metal-support interactions** (Tauster, S. J. et al. *Science* **1981**, 211, 1121–1125; Campbell, C. T. *Nat. Chem.* **2012**, 4, 597–598; van Deelen, T. W. et al. *Nat. Catal.* **2019**, 2, 955–970), **reducible oxide chemistry** (Wendt, S. et al. *Science* **2008**, 320, 1755–1759), **defect-mediated water activation** (Menetrey, M. et al. *Surf. Sci.* **2003**, 524, 49–62; Schaub, R. et al. *Phys. Rev. Lett.* 2001, 87, 266104-1–266104-4.), **Surface functionalization of the support** (Van Den Berg, R. et al. *ACS Catal.* **2015**, 5, 4439–4448), **hydroxyl effects** (Gunaseoriya, G. T. K. K. et al. *ACS Catal.* **2016**, 6, 3660–

3664; Saavedra, J. et al. *Science* **2014**, 345, 1599–1602; Parkinson, G. S. et al. *Nat. Mater.* **2013**, 12, 724–728.) and **kinetic and mechanistic studies of WGS on Pt/TiO₂** (Kalamaras, C. M. et al. *J. Catal.* **2009**, 64, 117–129; Azzam, K. G. et al. *Appl. Catal. B Environ.* **2008**, 80, 129–140) have been added or replaced in the revised manuscript. **These references have been added to the revised manuscript as 1, 4–7, 9–13, 20–23, and 32 (highlight in yellow).**

Comment 7: *I would also like to comment on a conceptual statement made in the rebuttal, which appears to carry over implicitly into the revised mechanistic framing. The authors state that “for gas-phase reactions involving two distinct reactant molecules, such as the water–gas shift reaction, catalysis at a single type of active site is often insufficient.” As written, this statement remains misleading and introduces conceptual ambiguities that has not yet been fully resolved in the revised text. The overwhelming majority of heterogeneous catalytic reactions involve two or more reactants, and yet in many well-established systems the dominant chemistry occurs on metal sites alone, even when multiple elementary steps and adsorbates are involved. Moreover, supported metal nanoparticles do not constitute a “single type of active site” in any strict sense, as they inherently present a distribution of terraces, edges, corners, and metal–support interfacial sites. While bifunctional or metal–support cooperative mechanisms certainly exist and are well documented for some reactions and materials, they are not the general rule, and invoking them requires careful evidence rather than broad generalization. I would therefore encourage the authors to revise the language in both the manuscript and the rebuttal to avoid implying that multi-reactant gas-phase catalysis generally requires chemically distinct active sites on different phases. A more accurate framing would acknowledge that metal–support cooperation is one possible mechanistic motif for WGS on reducible oxides, but not an inherent requirement dictated simply by the presence of multiple reactants.*

Response: We sincerely thank the reviewer for this professional correction, which has helped us clarify a significant conceptual ambiguity. In the revised manuscript, we have

completely corrected this erroneous statement. The manuscript now describes metal–oxygen vacancies synergy as one possible mechanism for the WGS reaction on reducible oxide–based catalysts. **The corresponding part was on Page 3, lines 6–7 in the revised manuscript (highlight in yellow).**

Comment 8: *Finally, the manuscript’s positioning still risks overselling novelty in a field where the coupled roles of hydroxyls, defect chemistry, and metal electronic structure in WGS have been discussed extensively. The work will be strongest if it is positioned around the specific and defensible contribution: a practical hydroxyl-regulation approach that yields a reproducible optimum and correlates with improved activity, with a mechanistic interpretation that is consistent with multiple probes but not asserted as uniquely proven where it is not. In particular, the manuscript should avoid implying that Pt L3-edge data under H₂ pretreatment definitively establishes Pt surface oxidation state under WGS conditions.*

Response: We sincerely thank the reviewer for the guidance on positioning our work. To avoid any overstatement of novelty, we have reframed the manuscript around the specific and defensible contribution as suggested.

1. Refocused abstract, discussion and the last first paragraph of introduction: We now position our primary contribution as the development of a practical and reproducible hydroxyl-regulation strategy via titanate precursor calcination that delivers an optimized and well-characterized catalyst. Supported by multiple characterizations, a mechanism interpretation is provided that rationalizes the role of hydroxyls in tuning the metal electronic state and the oxide surface structure. **The corresponding part was on Page 2, lines 1–14, Page 4, lines 17–26 and Page 19, lines 6–18 in the revised manuscript (highlight in yellow).**

2. Clarified mechanistic claims: Throughout the manuscript, we have replaced definitive language with cautious interpretations. Specifically, regarding the Pt L₃-edge data, we explicitly stated that the results obtained under H₂ atmosphere could not fully represent the exact state of Pt under WGS conditions, which could only imply a stronger

tendency for Pt species in 0.3Pt/(TiO₂-T-500) to be in the metallic state compared to that in 0.3Pt/(TiO₂-A). **The corresponding description was added on Page 9, lines 19–23 in the revised manuscript (highlight in yellow).**

***Comment 9:** Overall, the revision addresses several legitimate concerns and the performance improvements, including TOF-normalized differences as presented, are significant. I believe the work can be suitable for Nature Communications provided that the mechanistic claims are brought into tighter alignment with what is strictly demonstrated, and that either a discriminating experiment is added to strengthen the hydroxyl-to-vacancy-to-water-activation causality or the mechanistic framing is thoroughly revised to be appropriately cautious. In addition, the manuscript should preemptively address residual-ion effects from the synthesis route with quantitative elemental analysis or equivalent evidence.*

Response: We thank the reviewer for the positive assessment of our revisions and for the clear pathway to further improvement. In response to the reviewer's comments, we have made the following efforts to address the relevant concerns.

1. Mechanistic claims aligned with evidence: We have thoroughly revised the manuscript to bring all mechanistic interpretations into tighter alignment with the demonstrated evidence. Language implying causation has been replaced with more cautious phrasing (e.g., “suggests,” “is consistent with,” “points to”), particularly regarding the states of Pt and the role of vacancies. The proposed mechanism is now explicitly framed as a coherent model that best reconciles the collective data.

2. Conservative mechanistic framing: We have adopted the suggested path of a “thoroughly revised and appropriately cautious mechanistic framing.” The narrative no longer presents the hydroxyl–vacancy–water activation sequence as a proven causal chain, but as a self-consistent working model strongly supported by, and providing a rational explanation for, the full set of our observations.

3. Residual-ion effects addressed quantitatively: We combined evidence from ICP-MS quantification, XPS analysis, and thorough support washing ruling out the contribution of residual Na and Cl from synthesis to the observed activity enhancement.

Overall, we once again thank the reviewers for the professional and careful comments, which substantially improved the quality of our manuscript. We hope that our responses and revisions now meet the standards of *Nature Communications*.

Reviewer #2: *The authors have addressed all of my concerns, and I recommend the publication of this work in Nature Communications.*

Response: We thank the reviewer for the positive feedback on our work and for the time invested. Your expert comments have been instrumental in enhancing the quality of the manuscript.

Responses to the Reviewers' Comments and the Corresponding Revisions

Reviewer #1:

Comment 1: *The authors have now responded appropriately and substantively and have largely addressed my concerns. Small residual things remain. E.g., the manuscript still contains a few sentences that read more definitive than the new caveats warrant—for example, language like “predominant Pt0 species were present...” and “suggested... in the absence of size effects” is still stronger than “consistent with”, given their own acknowledgement that CO-band positions can shift with coverage/geometry - but I believe this is acceptable.*

Response: We sincerely thank the reviewers for their valuable time and professional comments, which has significantly enhanced the quality of our manuscript.